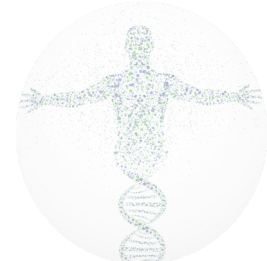




Ecole Thématique #2 Métabolomique et Santé Humaine

Aussois, 5 - 9 octobre 2026

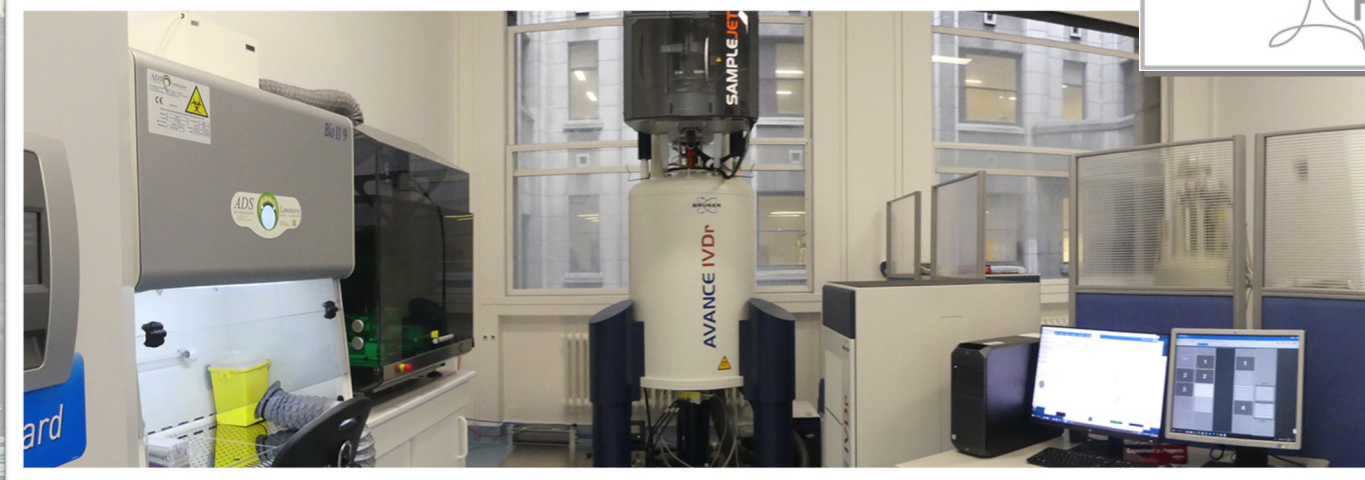


MetaboParis-Santé facility

www.metaboparis.fr



Gildas BERTHO



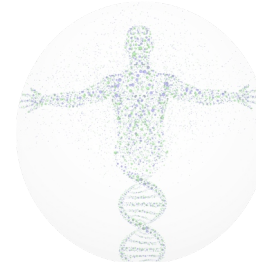


Ecole Thématique #2 Métabolomique et Santé Humaine

Aussois, 5 - 9 octobre 2026



SFBC
Société Française
de Biologie Clinique



NMR Session

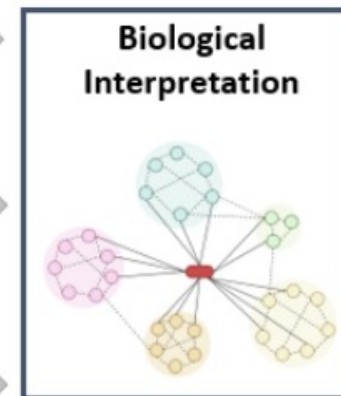
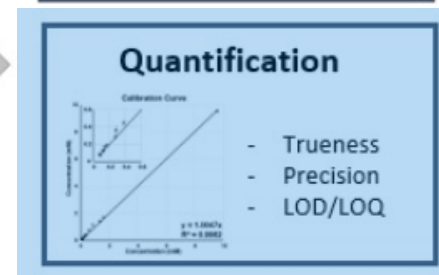
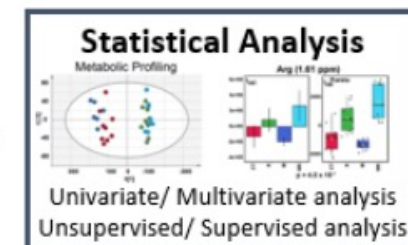
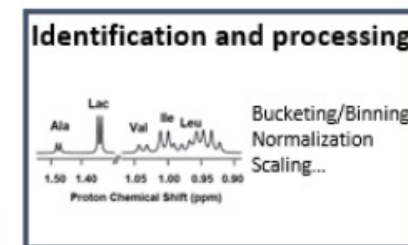
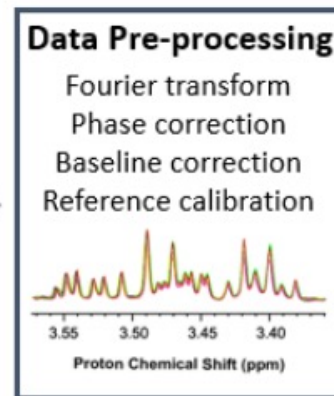
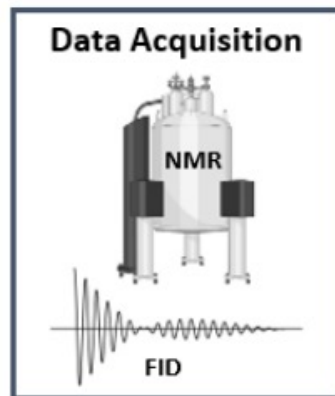
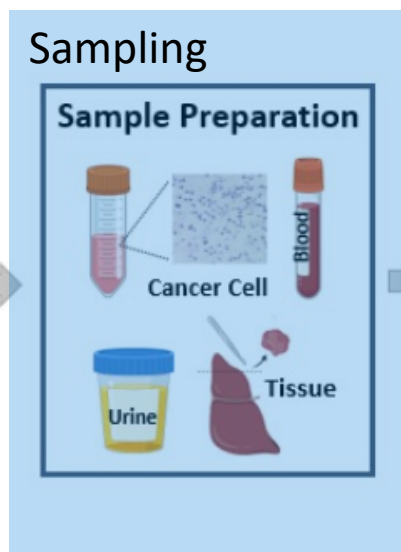
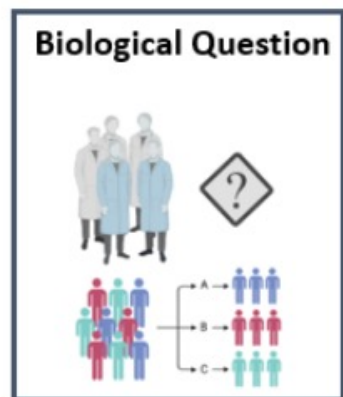
9:00

Gildas Bertho

10:45

Corentine Goossens

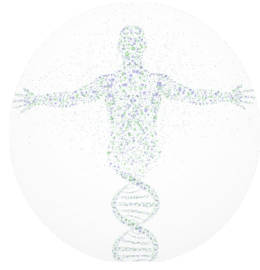
13:30 *Marine Letertre*
Pascal De Tullio



15:15

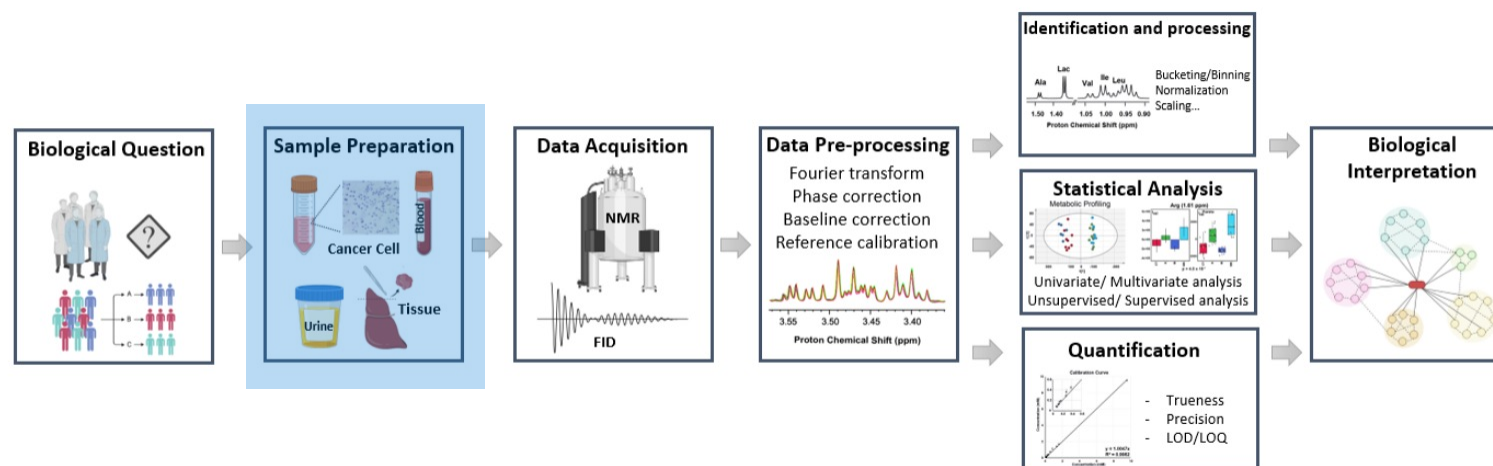
Corentine Goossens, Gildas Bertho

What are the Best Practices for Sampling and Sample Preparation?



Gildas BERTHO

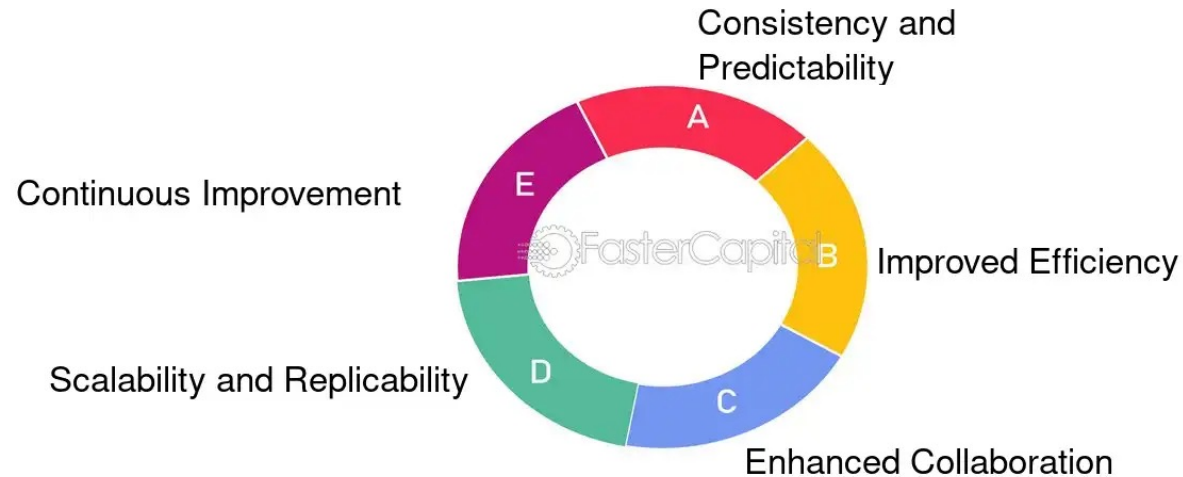
Recommended practices for the analysis of human cells, tissues, biofluids.



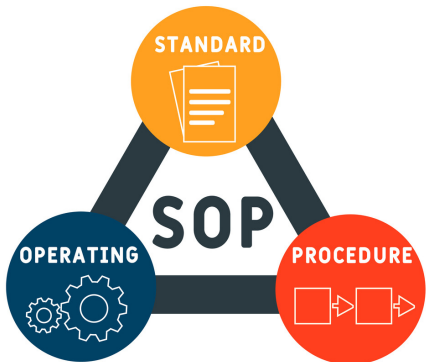
1) STANDARDISATION: WHY?

Benefits of Standardization for Project Delivery

- ✓ REPRODUCIBILITY INSIDE THE LAB
- ✓ INTERSITE ANALYSIS



Use SOPs (Standard Operating Procedures)



IT SEEMS TO BE RATHER SIMPLE TO DO

- ✓ ALWAYS USE THE SAME MATERIAL
- ✓ CONVINCE PEOPLE TO USE THE SAME MATERIAL ON DIFFERENT SITES



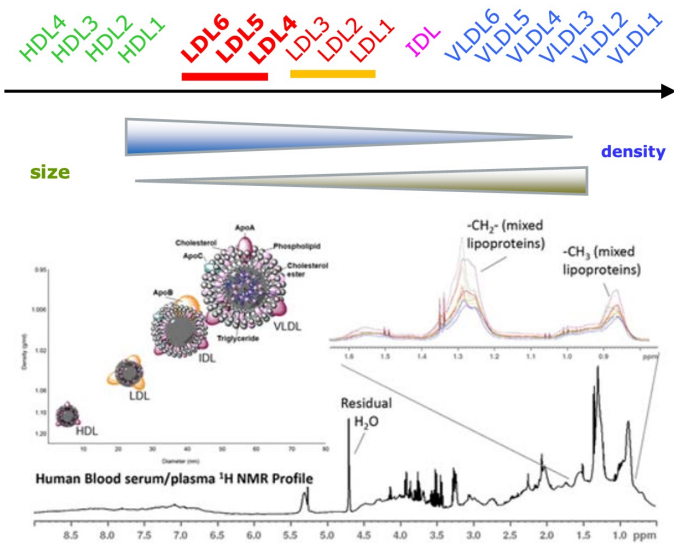
9 Benefits of Good SOPs

1. Saving Time
2. Decreasing Mistakes
3. Reducing Training Costs
4. Ensuring Consistent Results
5. Keeping People Safe
6. Empowering the Workforce
7. Supporting Quality Goals
8. Delegating Work
9. Increasing Compliance



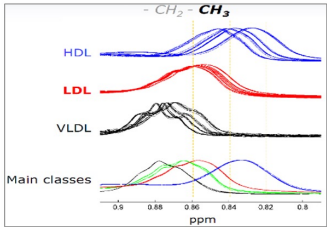
1) STANDARDISATION: WHY?

Example of benefits :
-Blood Analysis



115 Parameters:
CH, FC, TG, PL,
ApoA1,A2,B100

Main and
subfractions
VLDL, LDL, HDL



-Automatic Metabolite Identification & Quantification

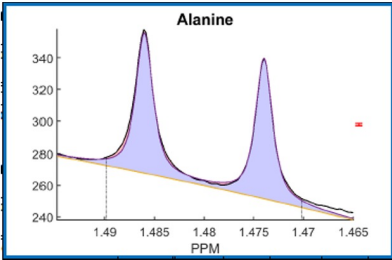
Automatic report

Bruker IVDr provides automatic identification on:

- 41 metabolites in Plasma/Serum
- 150 metabolites in Urine
- 115 Lipoprotein parameters in Plasma/Serum

3 Amino acids and derivatives

Compound	Conc. mmol/L	LOD mmol/L	r mmol/L	ρ %	Δ mmol/L	95% Range mmol/L	Graphics (*)
2-Aminobutyric acid	< 0.05	0.05	0.000	0	0.827	≤ 0.10	
Alanine	0.42	0.02	0.423	100	0.011	0.29 - 0.64	
Asparagine	< 0.05	0.05	0.000	0	4.734	≤ 0.08	
Creatine	0.02	0.01	0.024	100	0.002	≤ 0.07	
Creatinine	0.06	0.01	0.059	100	0.002	0.06 - 0.14	
Glutamic acid	0.11	0.05	0.106	25	0.083	≤ 0.24	
Glutamine	0.69	0.02	0.687	99	0.018	0.30 - 0.83	
Glycine	0.30	0.01	0.298	100	0.007	0.17 - 0.44	



-Lipoprotein report

Particle numbers

Total Concentration of ApoB carrying Particles

Key	Parameter	Value	Unit	95% Range of Model	Graphics (*)
TBPN	Total Particle Number	1777	nmol/L	876 - 2908	

(*) Gray horizontal boxes represent 95% range of model, black vertical lines represent sample value.

Lipoprotein Main Fractions

Key	Parameter	Value	Unit	95% Range of Model	Graphics (*)
VLPN	VLDL Particle Number	138	nmol/L	50 - 473	
IDPN	IDL Particle Number	98	nmol/L	36 - 316	
LDPN	LDL Particle Number	1452	nmol/L	760 - 2560	

(*) Gray horizontal boxes represent 95% range of model, black vertical lines represent sample value.

LDL Subfractions

Key	Parameter	Value	Unit	95% Range of Model	Graphics (*)
L1PN	LDL-1 Particle Number	232	nmol/L	98 - 567	
L2PN	LDL-2 Particle Number	268	nmol/L	47 - 427	
L3PN	LDL-3 Particle Number	282	nmol/L	51 - 499	
L4PN	LDL-4 Particle Number	300	nmol/L	77 - 577	
L5PN	LDL-5 Particle Number	248	nmol/L	86 - 615	
L6PN	LDL-6 Particle Number	182	nmol/L	91 - 815	

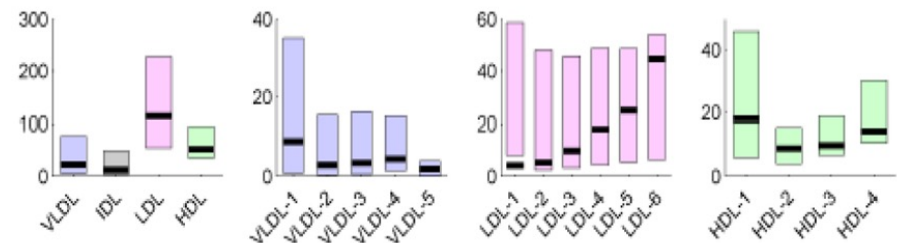
(*) Gray horizontal boxes represent 95% range of model, black vertical lines represent sample value.

Main class

Key	Parameter	Value	Unit	95% Range of Model	Graphics (*)
TPTG	TG	115	mg/dL	53 - 490	
TPCH	Chol	204	mg/dL	140 - 341	
LDCH	LDL-Chol	134	mg/dL	55 - 227	
HDCH	HDL-Chol	50	mg/dL	35 - 96	
TPA1	Apo-A1	146	mg/dL	112 - 217	
TPA2	Apo-A2	32	mg/dL	24 - 48	
TPAB	Apo-B100	98	mg/dL	48 - 160	

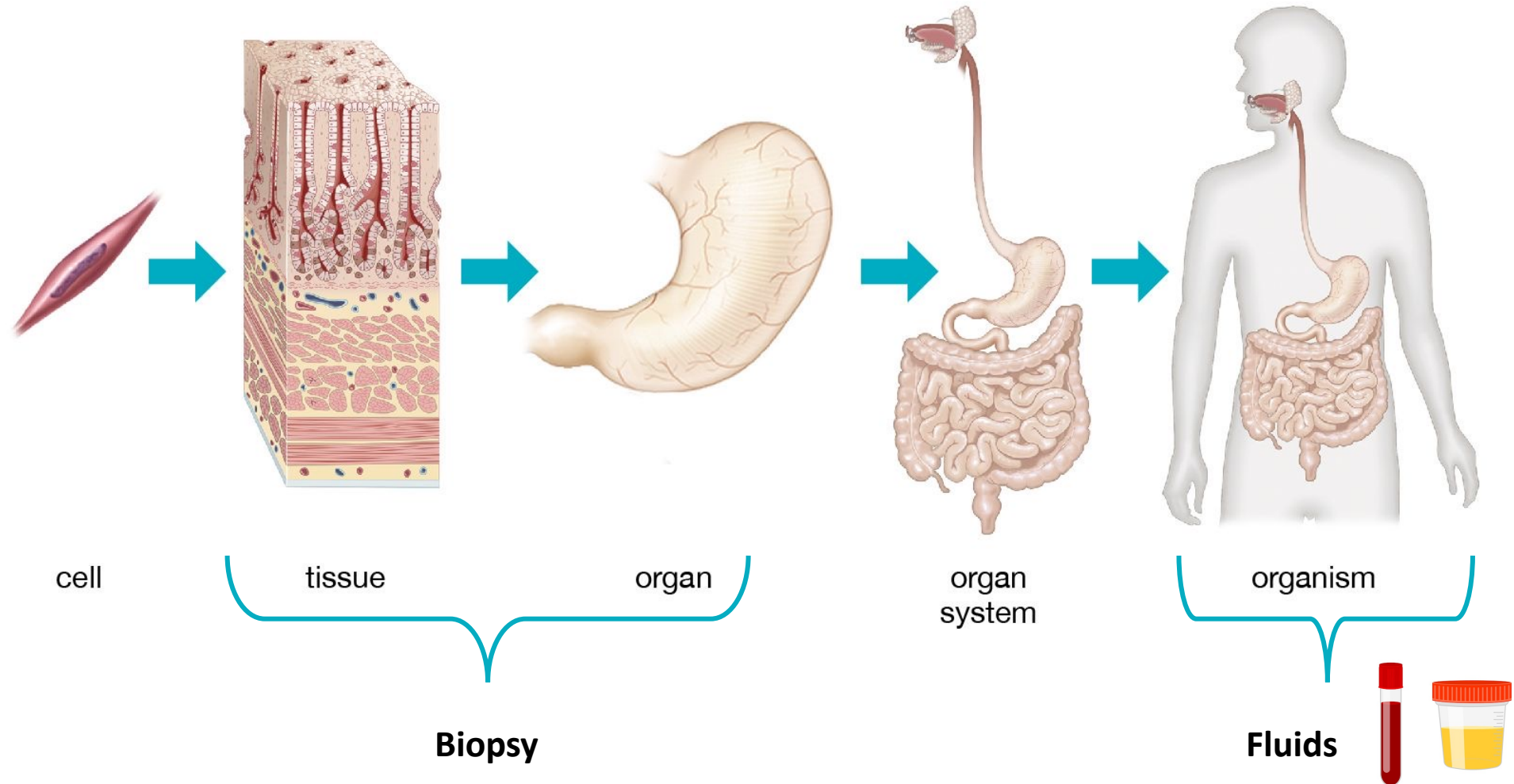
Subclass

Cholesterol distribution (concentrations in mg/dL together with 95% range of model)



From cells to organism

Levels of organization



Beckonert, O., Keun, H., Ebbels, T. *et al.* Metabolic profiling, metabolomic and metabonomic procedures for NMR spectroscopy of urine, plasma, serum and tissue extracts. *Nat Protoc* **2**, 2692–2703 (2007)

Cells



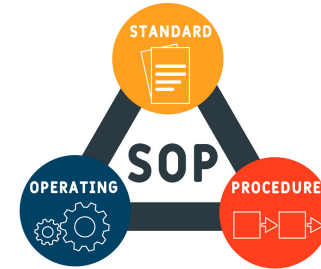
- Culture media & Cells



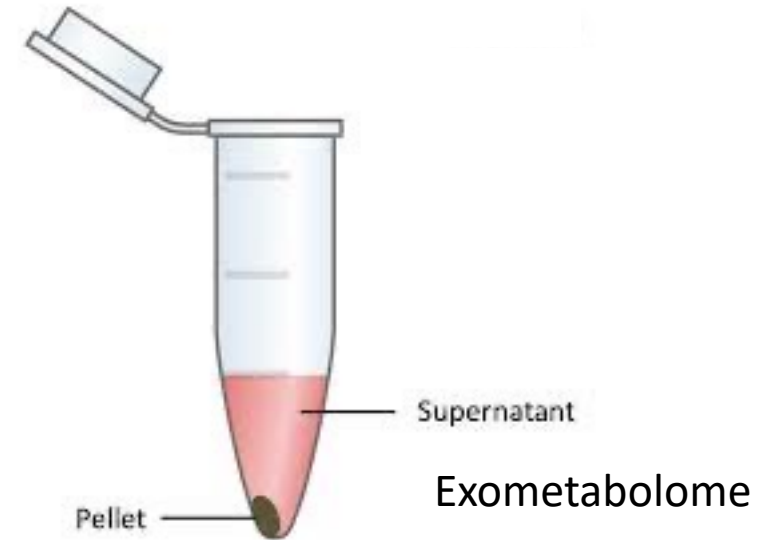
Be careful to the choice of :

- Culture media : HEPES, Tris, ...
- Glutamine or Glutamax™

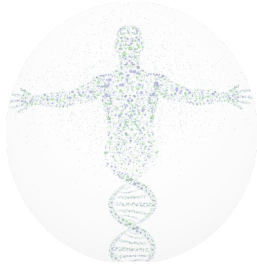
No SOPs



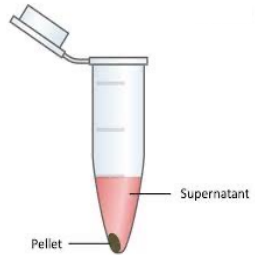
Centrifugation



Endometabolome



Supernatant



Extracellular medium (500uL minimum) out of -80°C for 30min



Take 432uL and transfer to an Eppendorf



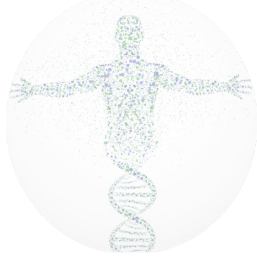
Add 168uL of **deuterated buffer** (see preparation in Supp. Mat.)



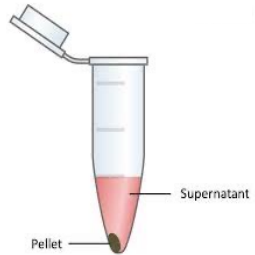
Remove everything once, avoiding bubbles to mix



Transfer to NMR tube



Cell Pellet



Extraction in $\text{CHCl}_3/\text{MeOH}/\text{H}_2\text{O}$

Approx. 10M frozen cells



Methanol/water mixture (0.4mL/0.065mL)



Supernatant recovery at 4°C (take 1mL cones with wide mouthpieces)



200μL of CHCl_3



Vortex



200μL CHCl_3 + 200μL H_2O



Vortex



15 min at 4°C (refrigerator)



Centrifugation 15 min at 4°C with 1K G

1

Upper phase
Aqueous phase

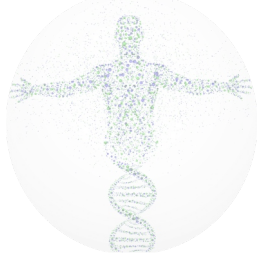
2

Centrifugation 10 min at 4°C with 12K G

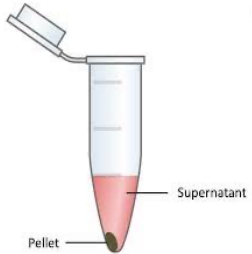


Lower Phase
Organic phase





Cell Pellet



Aqueous phase

Evaporation under nitrogen flux
(0.15 bar @ 18.5cm)

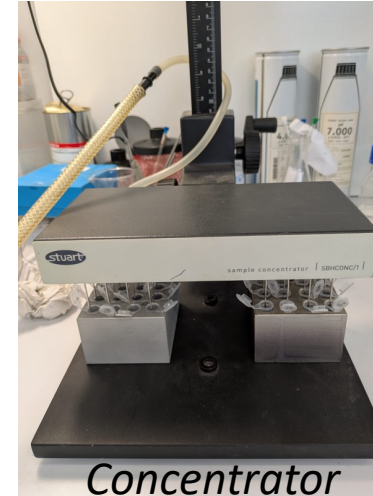
Pierce the caps with a needle
(5 Holes ; gauge 26)

Overnight water removal by
freeze-drying after freezing with N₂ (I)

Tip: Place the vertical tubes attached to each other with a rubber band

600μL D₂O + 0.005%TSP

600μL in the tube



Organic phase

Evaporation under nitrogen flux
(0.15 bar @ 18.5cm)

580μL CDCl₃ + 0.03% TMS

Vortex, then centrifugation
5 min @ 1K G

Recover everything except pellet

Analyses
RMN

Tissue, organe...



tissue



organ

30 to 40 mg of biopsy (wet mass)



Methanol/water mixture (0.4mL/0.065mL per biopsy)



2 Stainless Steel Beads (5 mm; qiagen 69989)



Grinding => 3 cycles of 45sec @ 25Hz (Wait 10 min in the ice between each run)



Recovery of the supernatant at 4°C (take a 1mL cone with the tip cut off)



200µL of CHCl₃



Vortex → 200µL CHCl₃ + 200µL H₂O → Vortex



15 min @ 4°C (fridge)



Centrifugation 15 min at 4°C @ 1K G



1

Upper phase
Aqueous phase

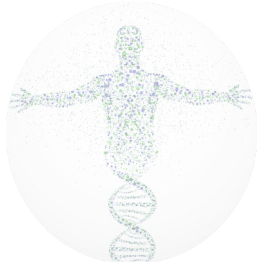
2

Centrifugation 10 min at 4°C @ 12K G

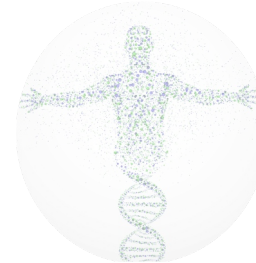
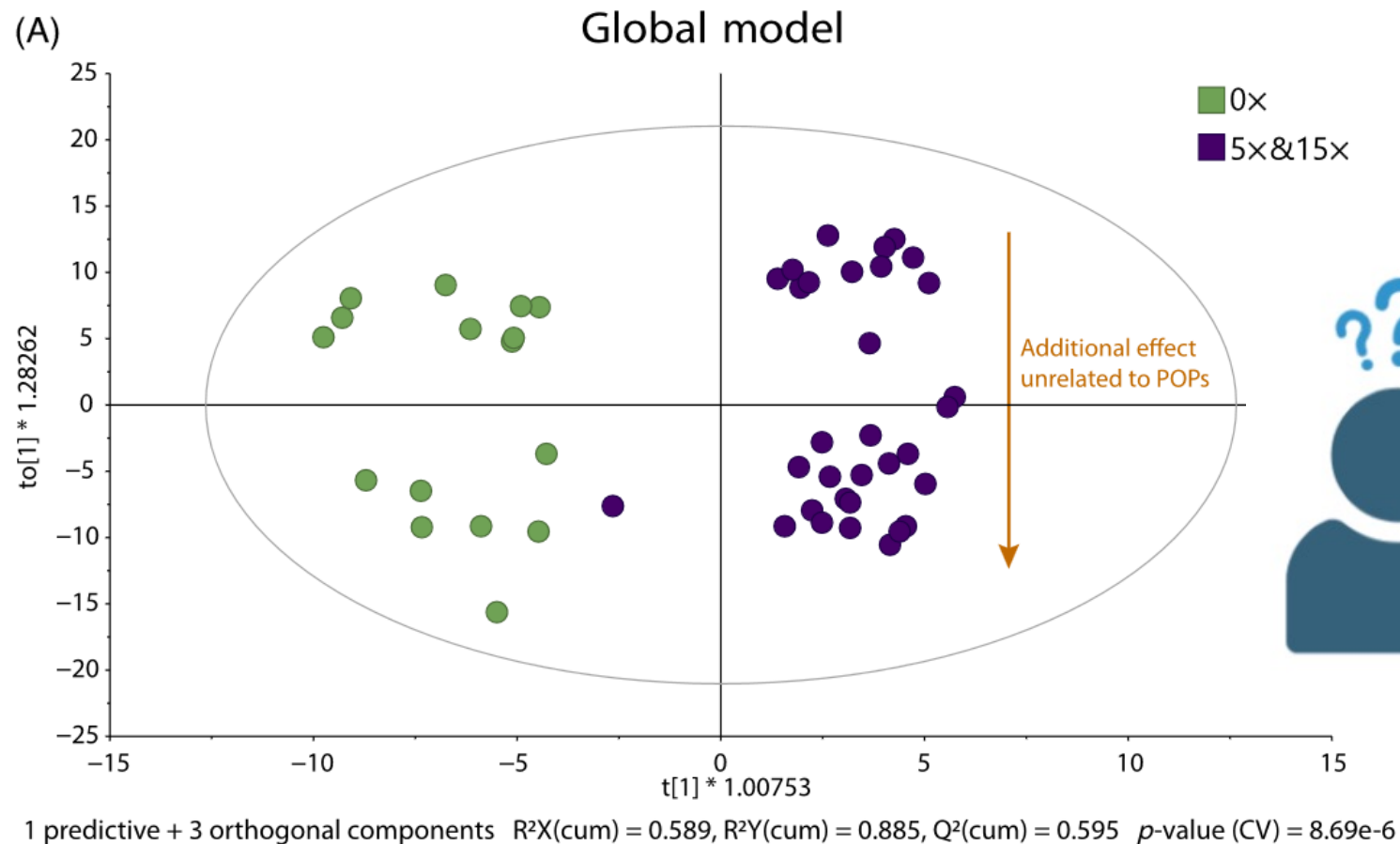
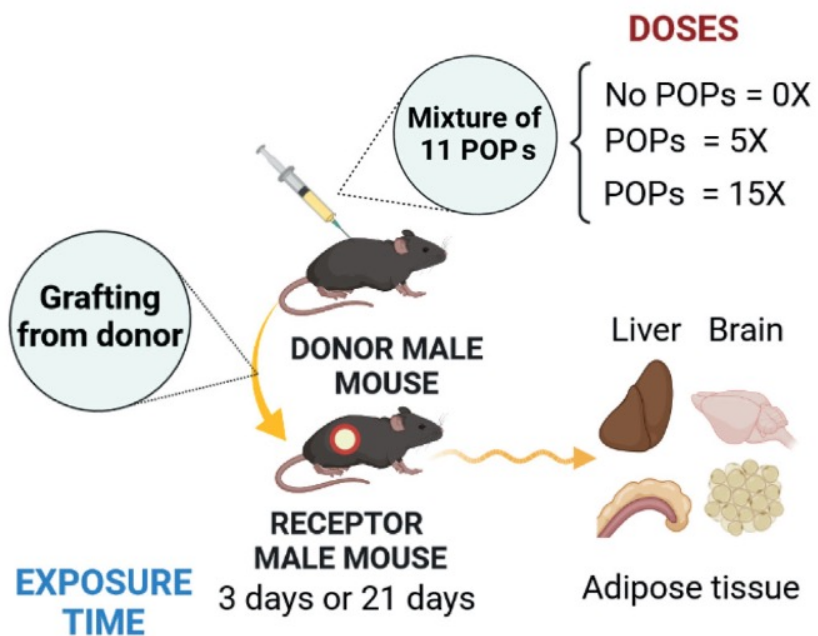


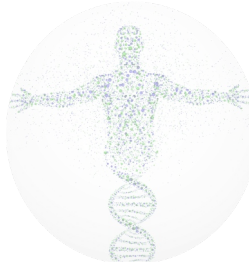
Lower Phase
Organic phase

Extraction in
CHCl₃/MeOH/H₂O



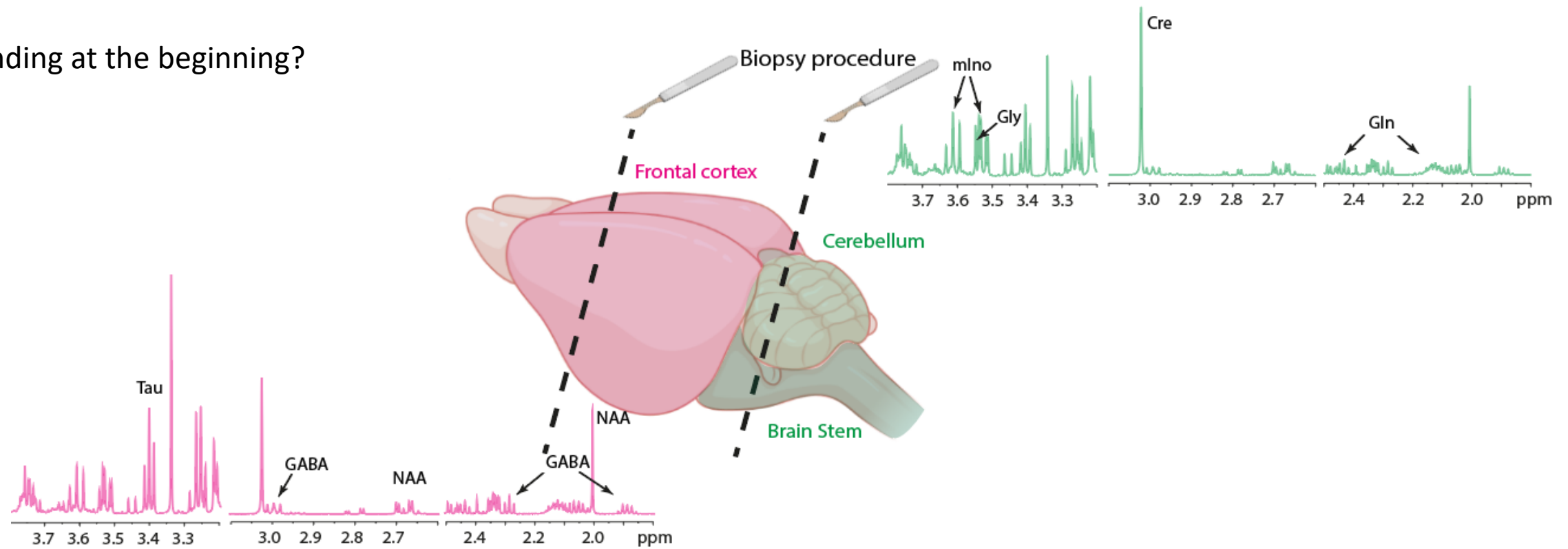
A problem found when studying organs



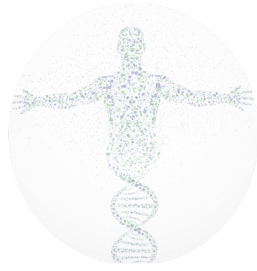


The problem of the heterogeneity of the organ!

Grinding at the beginning?



Biological fluids



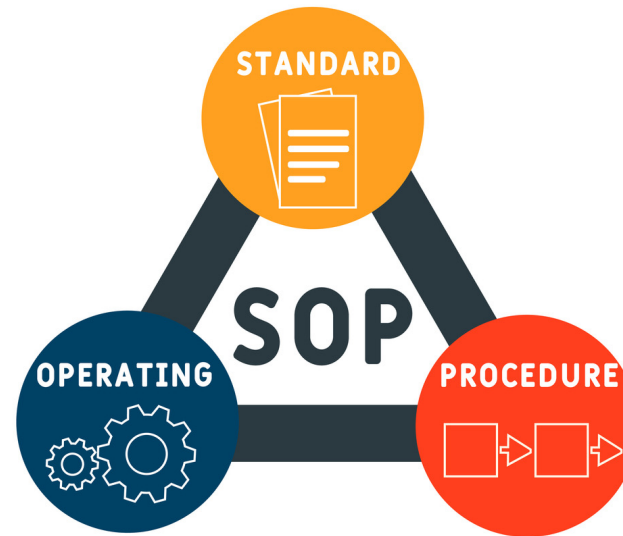
Label CE



Worldwide (Not CE)



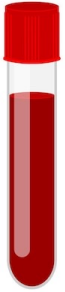
USA, label CE



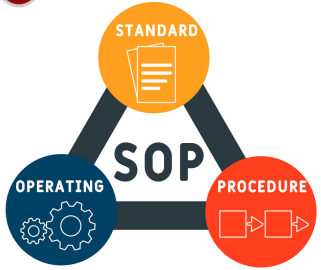
Recent reference for blood:

Recommendations for sample selection, collection and preparation for NMR-based metabolomics studies of blood. Metabolomics. 2025 May 10;21(3):66. doi: 10.1007/s11306-025-02259-7





SOPs Blood



Collection

- Blood collection in fasted state (e.g in the morning, 12 h fasted).
- Fresh EDTA-plasma into 6ml-tubes EDTA
(BD Ref. 367864, Vacutainer K2E 10.8 mg),
- Needle diameter 0.8mm.
- Gentle blood withdrawal, avoiding manipulation of the vein or a long time of tourniquet, which might result in hemolysis.
- Centrifugation at (6° C, 13 min at 1700 G).
- 1mL plasma per aliquot into **Cryovials 1.8 mL**
- Immediate deep freezing of plasma at -80° C.



Blood Samples

Summary

Aspect	Guidance	Recommendations
Sample Collection	No special procedures needed.	Great success with very old sample collections (deep-frozen for over 20 years).
Sample Preparation	Many factors (eating, exercise, stress, coffee, smoking, alcohol, medicines) can influence test results.	Proper preparation is crucial for sample and test quality.
Acceptable Samples	Standard clinical human venous fasting or non-fasting serum, or EDTA, citrate, or heparin plasma Samples.	Follow WHO guidelines on drawing blood: best practices in phlebotomy.
Sample Processing	Minimize time of blood and serum/plasma at room temperature.	Cool separated serum/plasma to +4°C as soon as possible.
Sample Storage	Deliver samples to Nightingale within three days from collection.	If delivery is delayed, freeze serum/plasma at -20°C, and further to -80°C (or colder) as soon as possible.
Long-term Storage	Long-term storage (>1 month) must be at -80°C or colder (e.g. liquid nitrogen).	Minimize freeze-thaw cycles (maximum of three recommended).
Sample Volume	500 µL classically ; 100 µL is now possible	Ask more volume (e.g. 1 mL) to be sure to have enough
Sample Container	Outer diameter of tube must be less than 13 mm.	Cryovials 1.8 mL with Septum (Bruker Store) are recommended



Blood Samples

How blood samples need to be collected for our analysis ?

No special procedures are needed. Great success studies even very old sample collections (deep-frozen for over 20 years).

- Advice on **sample collection**

Many factors occurring before the collection of Samples – such as eating, physical exercise, stress, coffee, smoking, alcohol and medicines (including natural products, vitamins, nutritional supplements) – may influence the results of laboratory tests. Proper preparation of patients for blood sampling is of vital significance for the quality of the Sample and test results. It is recommended to consult Nightingale in case of untypical non-populationbased studies, for example in the case of interventional studies, to test/pilot potential effects of interventions before performing extensive studies. For the Service, standard clinical human venous fasting or non-fasting serum, or EDTA, citrate, or heparin plasma Samples are acceptable. Nightingale recommends to follow for example the WHO guidelines on drawing blood: best practices in phlebotomy on applicable parts. The Service is compatible with all standard clinical blood samples collections. It is recommended to minimise the time of the blood as well as serum/plasma at the room temperature. The separated serum/plasma samples should be cooled down to +4°C as soon as possible. Customer shall follow the blood collection tube manufacturer's instructions for serum/plasma separation.

(?) For blood-drawing practices we recommend to follow for example [the official WHO guidelines on drawing blood: best practices in phlebotomy](#). Additionally, the [Joint EFLM-COLABIOCLI Recommendation for venous blood sampling](#) (Simundic et al., 2018) provides a clear step-by-step guide for blood sampling. If the conditions of fieldwork don't allow for rapid sample processing (i.e., serum/plasma separation) after blood sampling, it is recommended to store the blood samples at +4°C until they are processed, as this will slow down sample degradation processes such as glycolysis. It is recommended to process the blood samples as soon as possible, optimally within the same day of sample collection. It is primarily important to follow blood collection tube manufacturer's instructions and details such as precise clotting time and centrifugation settings.

- Advice on **sample storing**

The Samples shall be delivered to Nightingale within three days from sample collection, as advised by the standard practice in clinical chemistry. If the Samples are not delivered within the said time, the serum/plasma must be frozen at -20°C and, if possible, further to -80°C (or colder, e.g. liquid nitrogen) as soon as possible (preferably on the day of collection). Long-term storage (>1 month) of the Samples must always be -80°C or colder (e.g. liquid nitrogen).

(?) Once the blood samples have been processed and the separated serum/plasma has been transferred to a secondary tube, it is generally suggested to rapidly deep-freeze the separated serum/plasma for long-term storage. It would be optimal to freeze the serum/plasma samples at -80°C or colder as soon as possible, preferably on the day of collection and processing, and keep them at these temperatures for long-term storage. If the serum/plasma samples cannot be stored in a deep-freezer immediately after the separation, you can store them at +4°C for up to three days, or at -20°C for up to one month. Additionally, freeze-thaw cycles should be minimized. Based on Nightingale's experience after three freeze-thaw cycles, the biological quality of the samples starts to deteriorate.

- Advice on **sample volume**

500 µL, possible 100 µL

- **Sample container**

Several types of tubes and containers are accepted. The outer diameter of a tube must be less than 13 mm. Screw cap microtubes (0.5 ml, e.g., Sarstedt 72.730.005; or 2 ml, e.g., Sarstedt 72.694.006) are the most convenient ones.



Urine Sample

Summary

Aspect	Guidelines	Details
Sample Collection	Use first morning (or 24-hour collection)	Proper patient preparation is crucial for accurate test results
	Use a collection system without additives or preservatives	Minimize the time urine spends at room temperature
Sample Storage	Freeze samples at -20°C as soon as possible	Long-term storage (>1 month) should be at -80°C or colder
Sample Volume	Aim for a minimum sample volume of 500 μL .	Sending larger volumes (e.g. 600 μL) ensures accurate aspiration
Sample Container	Use tubes with an outer diameter less than 13 mm	Screw cap microtubes (0.5 ml or 2 ml) are recommended
Shipping	Refer to shipping checklist	Follow guidelines for safe and proper sample handling and shipping



Urine Sample

- Advice on **sample collection**

Many factors occurring before the collection of Samples – such as eating, physical exercise, stress, coffee, smoking, alcohol and medicines (including natural products, vitamins, nutritional supplements) – may influence the results of laboratory tests. Proper preparation of patients for urine sampling is of vital significance for the quality of the Sample and test results. For the Service, first morning urine or 24-hour urine collection is recommended. A collection system with no additives or preservatives should be used. It is recommended to minimise the time of the urine at room temperature.

- Advice on **sample storing**

The Samples must be frozen at -20°C and, if possible, further to -80°C (or colder, e.g. liquid nitrogen) as soon as possible (preferably on the day of collection). Long-term storage (>1 month) of the Samples should always be -80°C or colder (e.g. liquid nitrogen).

- Advice on **sample volume**

Absolute minimum sample volume is 500 µL of urine. However, to ensure that we are able to aspirate required volume for your vials, we recommend to send a larger volume, for example 600 µL or even more. This allows also accurate aspiration of urine if, for some reason, there is some precipitate (e.g. protein) in the urine sample. The required volume per Sample shall be delivered in a single tube.

- **Sample container**

Several types of tubes and containers are accepted. The outer diameter of a tube must be less than 13 mm. Screw cap microtubes (0.5 ml, e.g., Sarstedt 72.730.005; or 2 ml, e.g., Sarstedt 72.694.006) are the most convenient ones.

- Advice on **sample shipping**

For advice on **sample handling and shipping**, take a look at our [checklist](#).

➤ Sample Identification

Each sample must be uniquely identified with a barcode or matrix code and packed in an orderly fashion. The identification code, sent in table format alongside the samples, must match the code on the sample.



➤ Shipping Conditions

Frozen samples must be shipped with dry ice and a temperature monitor. Customer must confirm shipment schedule, obtain clearance, and provide a tracking code.



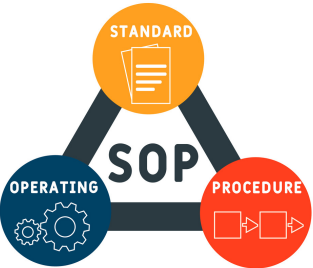
➤ Packing Instructions

Packaging must be strong, leak-proof, and include a primary receptacle, secondary packaging, and absorbent material. The outer packaging must be labeled “BIOLOGICAL SUBSTANCE, CATEGORY B” and “UN 3373”.



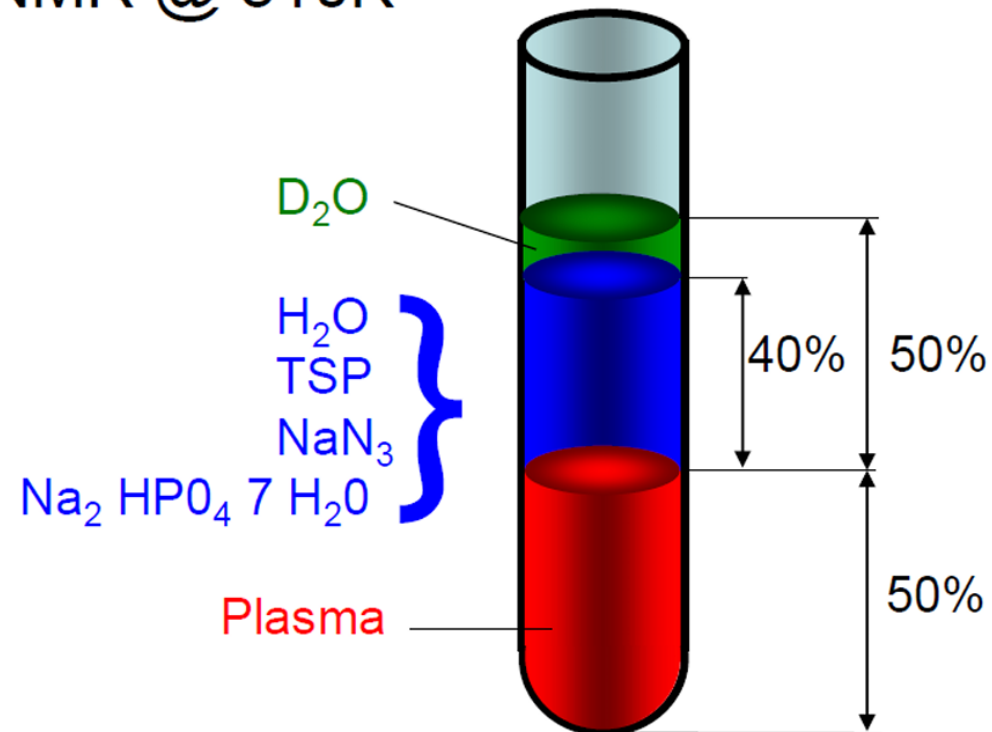
SOPs Blood

Preparation Procedure – Plasma/Serum



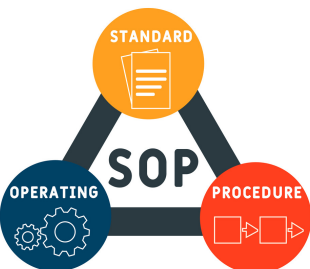
- Carefully thaw the samples at room temperature for ca. 30min.
- Add 400µL of Bruker plasma buffer.
- Add 400µL of plasma/serum into a Cryovial / Eppendorf.
- Shake the mixture gently for 1min (do NOT use the Vortex mixer).
- Transfer 600µL of well mixed sample into a 5 mm 7" NMR tube or a 5 mm SampleJet rack tube.

NMR @ 310K



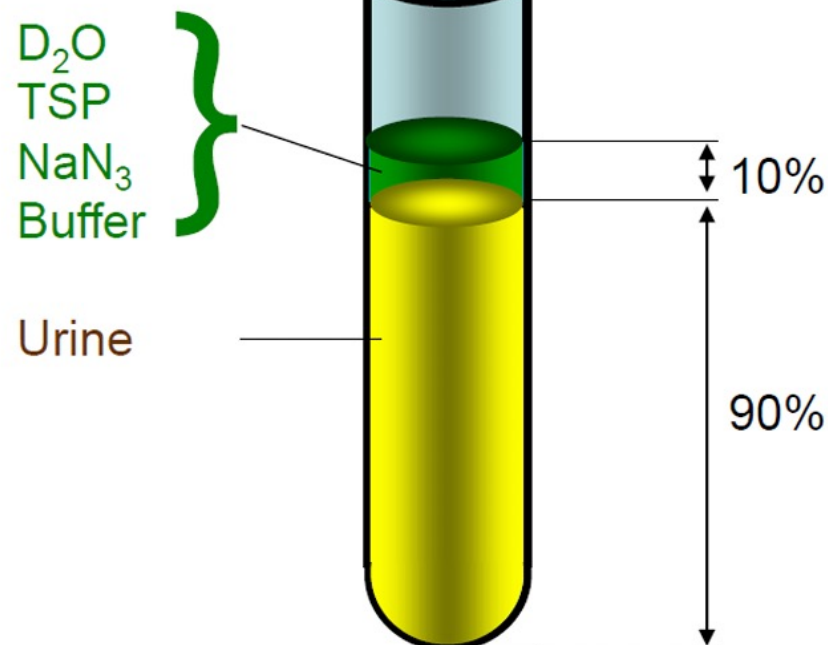
SOPs urine

Preparation Procedure



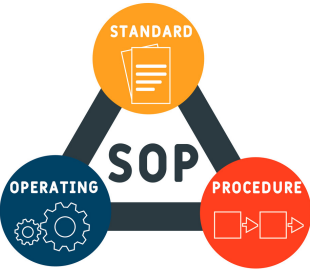
- Carefully thaw the samples at room temperature for ca.30 min.
- (Centrifuge the samples for 5 to 10min at ~2000 RCF).
- Add 100 μ L of Bruker urine buffer.
- Add 900 μ L of urine into a Cryovial / Eppendorf
- Mix the buffered urine for 30 seconds on the Vortex mixer.
- Transfer 600 μ L of well mixed sample into a 5mm 7" NMR tube or a 5mm SampleJet rack tube.

NMR @ 300K



Shake strongly for mixing!

SOPs in Calibration and Measurement



• Software

- Automation software scripts
- Experimental parametersets
- ICON NMR configuration
- SampleTrack configuration

• Reference Samples and Buffers

- Calibration samples for daily Quality Control and Optimization
- Pseudo Bodyfluid test samples
- Buffer solutions (starter kit)

• Sample preparation SOPs / Sample NMR Measurements SOPs

• Quality Control SOPs

FILCOR_NOESY FILCOR_ZGPR_O1 IVDr_CPMG IVDr_DAS IVDr_DIFF IVDr_GRADPROF IVDr_IRES IVDr_NOESY IVDr_ZG30 IVDr_ZGPR IVDr_ZGPR_O1 IVDr_ZGPRZD MEOD_TEMPAL_300K MEOD_TEMPAL_300K_3mm MEOD_TEMPAL_310K MEOD_TEMPAL_310K_3mm PROF_CSF_IRES PROF_CSF_IRES_3mm PROF_CSF_NOESY PROF_CSF_NOESY_3mm PROF_CSF_ZGPR PROF_CSF_ZGPR_3mm PROF_CSF_ZGPR_O1 PROF_CSF_ZGPR_O1_3mm PROF_MEQH_IRES PROF_MEQH_IRES_3mm PROF_MEQH_NOESY PROF_MEQH_NOESY_3mm PROF_MEQH_ZG30 PROF_MEQH_ZG30_3mm PROF_MEQH_ZGPS PROF_MEQH_ZGPS_3mm	PROF_PLASMA_CPMG PROF_PLASMA_CPMG_3mm PROF_PLASMA_DAS_A PROF_PLASMA_DAS_L PROF_PLASMA_DIFF PROF_PLASMA_DIFF_3mm PROF_PLASMA_IRES PROF_PLASMA_IRES_3mm PROF_PLASMA_NOESY PROF_PLASMA_NOESY_3mm PROF_PLASMA_ZGPR PROF_PLASMA_ZGPR_3mm PROF_PLASMA_ZGPR_O1 PROF_PLASMA_ZGPR_O1_3mm PROF_URINE_DAS_A PROF_URINE_DAS_N PROF_URINE_IRES PROF_URINE_IRES_3mm PROF_URINE_NOESY PROF_URINE_NOESY_3mm PROF_URINE_ZGPR PROF_URINE_ZGPR_3mm PROF_URINE_ZGPR_O1 PROF_URINE_ZGPR_O1_3mm QUANTREF600C_BA QUANTREF600C_BA_3mm QUANTREF600C_BA_310K QUANTREF600C_BA_310K_3mm QUANTREF600C_BA_O1 QUANTREF600C_BA_O1_3mm QUANTREF600C_BA_O1_310K QUANTREF600C_BA_O1_310K_3mm	QUANTREF600C_CS QUANTREF600C_CS_3mm QUANTREF600C_CS_O1 QUANTREF600C_CS_O1_3mm QUANTREF600C_PS QUANTREF600C_PS_3mm QUANTREF600C_PS_O1 QUANTREF600C_PS_O1_3mm QUANTREF600C_UR QUANTREF600C_UR_3mm QUANTREF600C_UR_O1 QUANTREF600C_UR_O1_3mm SUC_GRADPROF SUC_GRADPROF_3mm SUC_NOESY SUC_NOESY_3mm SUC_NOESY_310K SUC_NOESY_310K_3mm SUC_ZGPR SUC_ZGPR_3mm SUC_ZGPR_310K SUC_ZGPR_310K_3mm SUC_ZGPR_O1 SUC_ZGPR_O1_3mm SUC_ZGPR_O1_310K SUC_ZGPR_O1_310K_3mm SUC_ZGPRZD SUC_ZGPRZD_3mm SUC_ZGPRZD_310K SUC_ZGPRZD_310K_3mm
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IVDr Sample Kit



IVDr Buffers

How to know that samples are good?

- **What is the history of samples, from collection to analyze?**
- Has the protocol been well followed?
- Are the old samples still good?
- ...

1 Sample Preparation & Handling

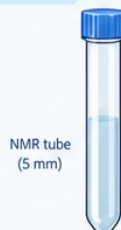
Quality Control in NMR Analysis

Reliable data • Accurate results • Confident decisions

1 Sample Preparation & Handling

Good sample quality is the foundation of reliable NMR data.

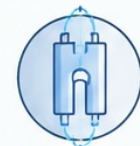
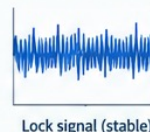
- ✓ Correct solvent (e.g. CDCl_3 , D_2O)
- ✓ Appropriate concentration
- ✓ Clean, dry and de-gassed (when needed)
- ✓ Proper tube and high-quality cap



2 Instrument Performance Check

Verify that the spectrometer is working within specifications.

- ✓ Magnet stability (field homogeneity)
- ✓ Shim quality
- ✓ Temperature control
- ✓ Tuning and matching
- ✓ Lock signal (deuterium)

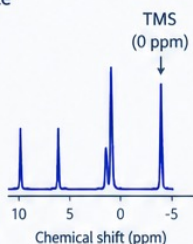


Field homogeneity (Shim)

3 Reference & Calibration

Use standards to ensure accurate chemical shifts, sensitivity and quantification.

- ✓ Chemical shift reference (e.g. TMS, DSS)
- ✓ Internal standard (if needed)
- ✓ Pulse calibration (90° pulse)
- ✓ Frequency and phase calibration

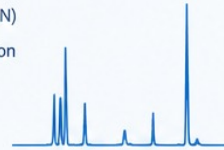


Quality control ensures your NMR data is accurate, reproducible and fit for purpose.

4 Data Quality Assessment

Evaluate the spectrum and processing for consistent, high-quality data.

- ✓ Signal-to-noise ratio (S/N)
- ✓ Line width and resolution
- ✓ Baseline stability
- ✓ Phase correction
- ✓ Absence of artifacts

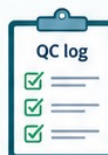


Good quality spectrum

5 Ongoing Monitoring & Documentation

Track performance over time and keep records for traceability and compliance.

- ✓ Use QC samples (e.g. standard mixture)
- ✓ Monitor key parameters (S/N, chemical shift, etc.)
- ✓ Maintain logs and instrument records
- ✓ Follow SOPs and regulatory requirements (e.g. GMP, ISO)



Accurate identification & quantification



Reproducible results



Regulatory compliance



Better science, stronger outcomes

NMR is :

- ✓ Robust
- ✓ Quantitative

Test of the Quality for Pre-Analysis and Analysis

Blood

- Spectral Quality
- Preparation Quality
- Matrix Identity
- Matrix Integrity
- Matrix Contamination

Urine

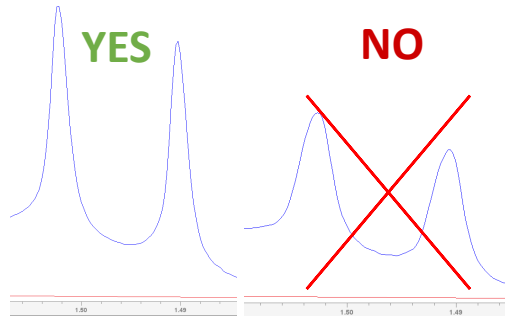
- Spectral Quality
- Matrix Identity
- Matrix Integrity
- Matrix Contamination
- Medication Test
- Further Indicative Parameter

Quality test for Blood

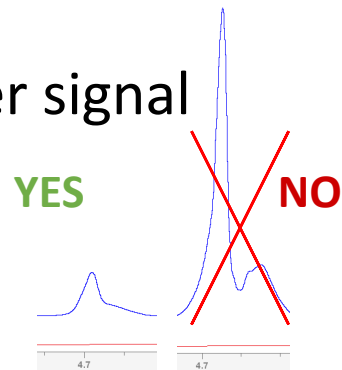
Blood – Spectral Quality

Parameters

- Linewidth at the Alanine doublet

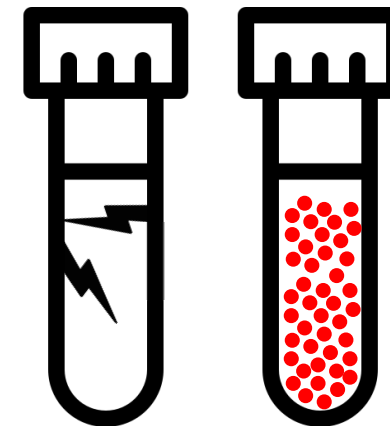


- Residual water signal



Signification

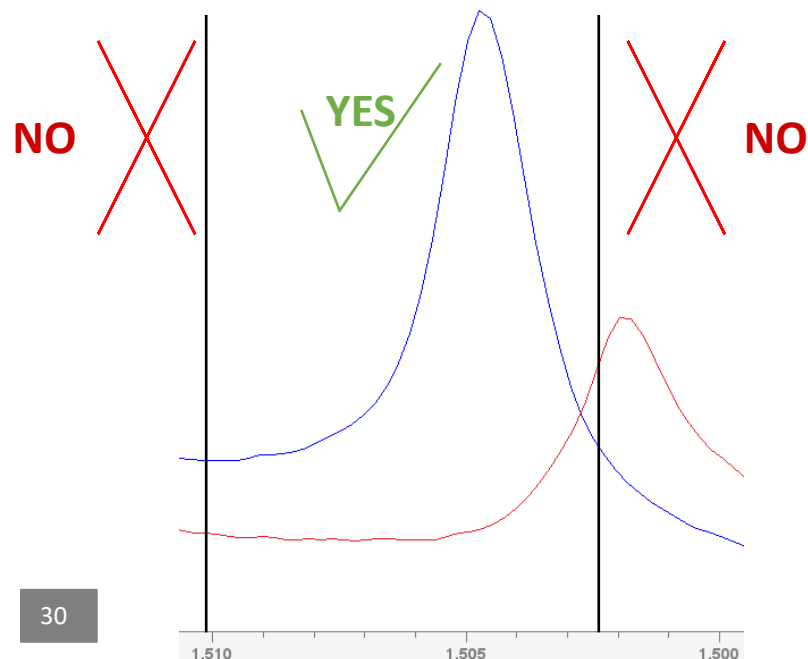
- Bad shim :
 - Tube Imperfection
 - Particles in the sample
 - Non homogeneous sample



Blood – Preparation Quality

Parameters

- Shift in ppm at the left peak of Alanine doublet



Signification

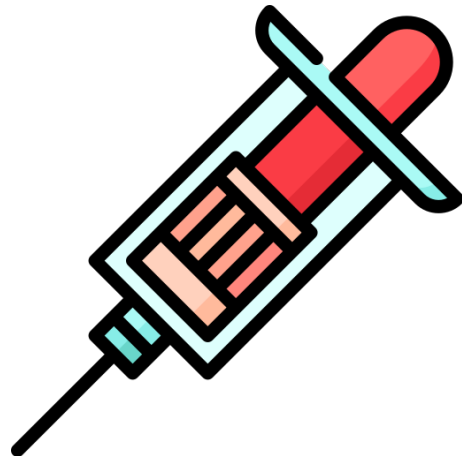
- Bad ratio sample/buffer at the preparation



Blood – Matrix Identity

Parameters

- Concentration of :
 - **K-EDTA**
 - **Ca-EDTA**
 - **Citrate**
- } > 3 mmol/L



Signification

- Plasma (EDTA, Citrate) or Serum (Plasma Heparine)
- Wrong sample collection

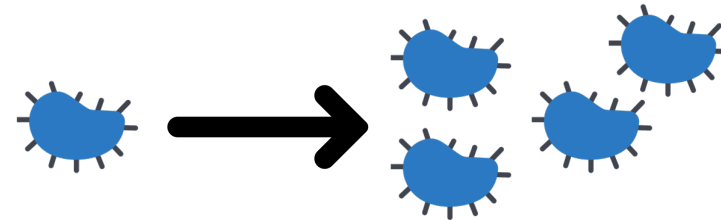
Blood – Matrix Integrity

Parameters

- Concentration and/or shift of :
 - Acetic acid** > 0.1 mmol/L
 - Citric acid-B** > 3 mmol/L
 - Formic acid** > 0.2 mmol/L
 - D-Glucose-alpha** > 3.2 mmol/L
 - Lactic acid** > 8.1 mmol/L
- > 0.002 ppm

Signification

- Enzymatic activities or even bacterial growth, because :
 - Inappropriate temperature exposure**
 - Inappropriate pre-analytics handling**



Metabolic Signatures of Hemolysis

NMR markers	Trend in Hemolysis	Origin
Lactate	↑	High intracellular concentration in erythrocytes
Alanine	↑	Release of cellular content
Reduced Glutathion (GSH)	↑↑	Highly abundant in red blood cells
ATP/ADP/AMP (when detectable)	↑	Intracellular content
Hypoxanthine	↑	Nucleotide degradation following cell lysis
Formate	↑ in some cases	Altered cellular metabolism
Glucose	Variable, often ↓	Residual cellular consumption
Pyruvate	Variable	Disturbance of energy metabolism

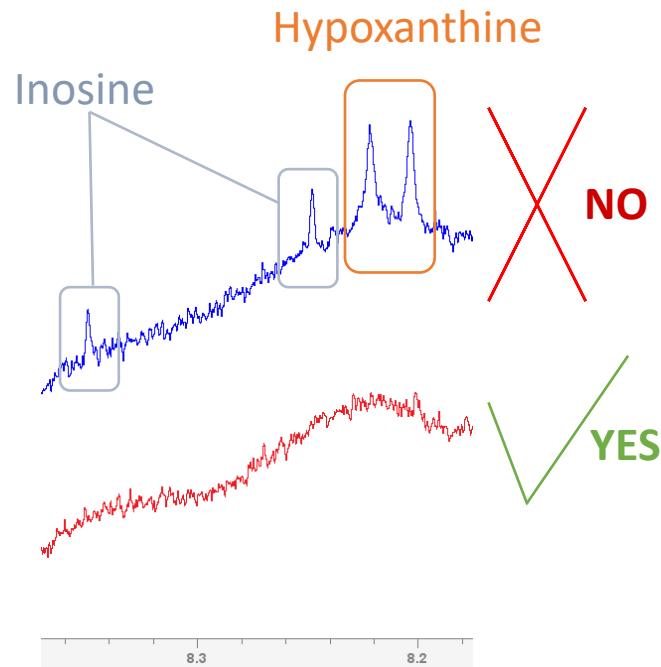
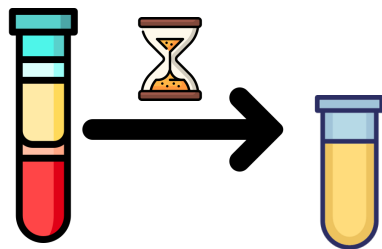
Blood – Matrix Integrity (continued)

Parameters

- Presence of :
 - Inosine
 - Hypoxanthine

Signification

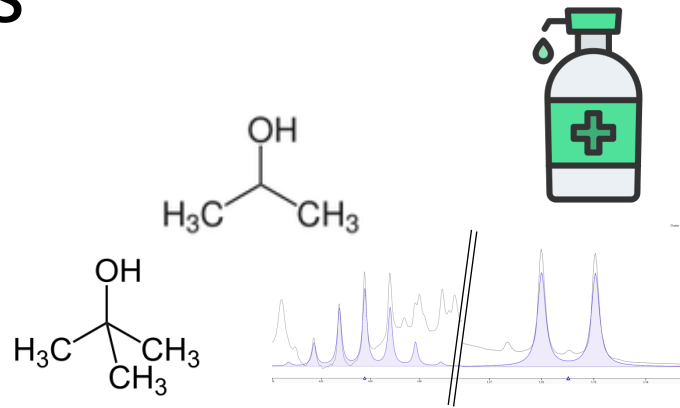
- Red blood cells not immediately separated :
 - Inappropriate temperature exposure
 - Inappropriate pre-analytics handling



Blood – Matrix Contamination

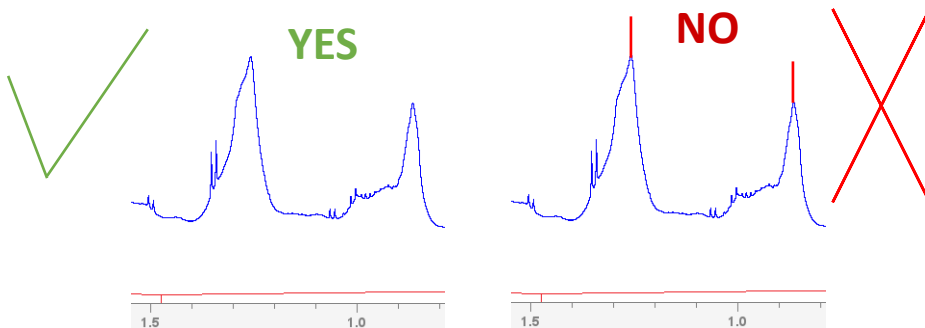
Parameters

- Presence of :
 - **Isopropanol**
 - **Tert-Butanol**
- Presence of :
 - **Sharp peaks at 0.8ppm and 1.25ppm at the top of lipoprotein signals**



Signification

- Solvent used at many step of pre-analytical procedures
 - **Antiseptic and disinfectant**
 - **Fume hood and benchtop cleaning**
- Low quality of cryovial used (presence of polymers in sample)

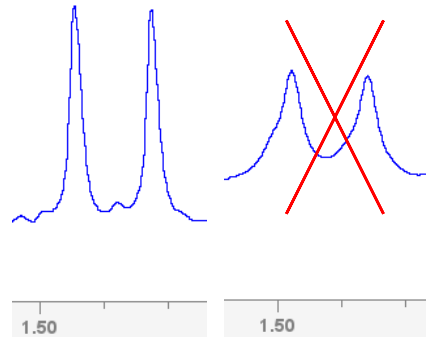


Quality tests for Urine

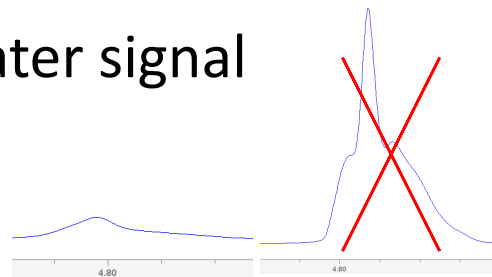
Urine – Spectral Quality

Parameters

- Linewidth at the Alanine doublet

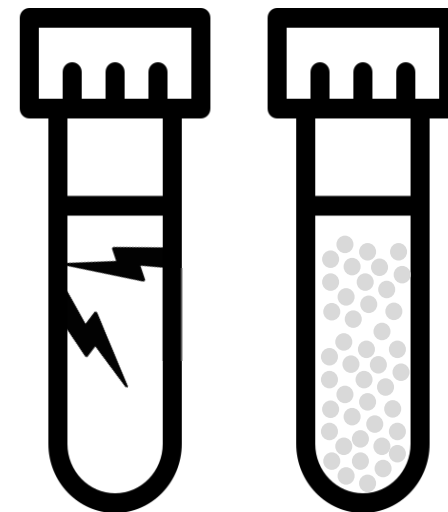


- Residual water signal



Signification

- Bad shim :
 - Tube Imperfection
 - Particles in the sample
 - Non homogene sample



Urine – Matrix Identity

Parameters

- Concentration of :
 - **Creatinine** 0.6 - 29 mmol/L
 - **Alanine** > 2.2 mmol/L
 - **Fumaric acid** > 0.07 mmol/L

Signification

- Wrong sample collection



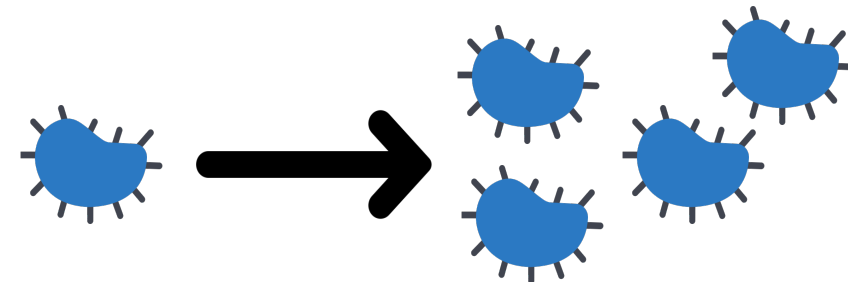
Urine – Matrix Integrity

Parameters

- Concentration of :
 - **Acetic acid** > 2.9 mmol/L
 - **Benzoic acid** > 0.45 mmol/L
 - **Formic acid** > 0.71 mmol/L
 - **Hippuric acid** > 15.1 mmol/L
 - **Lactic acid** > 2 mmol/L
 - **Succinic acid** > 0.44 mmol/L
 - **Citric acid** Non presence

Signification

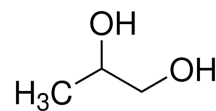
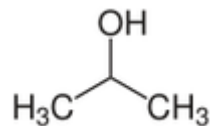
- Bacterial growth, because :
 - **Inappropriate temperature exposure**
 - **Inappropriate pre-analytics handling**



Urine – Matrix Contamination

Parameters

- Presence of :
 - **Isopropanol**
 - **Propylene Glycol**



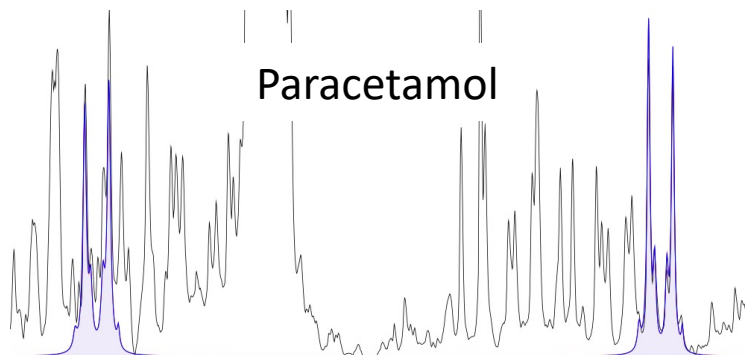
Signification

- Solvent used at many step of pre-analytical procedures

Urine – Medication Test

Parameters

- Presence of :
 - **D-Mannitol** > 0.15 mmol/L
 - **D-mannose-alpha** > 0.06 mmol/L
 - **Paracetamol** > 0.06 mmol/L
 - **Paracetamol-glucuronide** > 0.06 mmol/L



Signification

- Patient medication



Urine – Further Indicative Parameters

Parameters

- Concentration of :
 - **3-Hydroxybutyric acid** > 0.24 mmol/L
 - **Acetone** > 0.04 mmol/L
 - **Acetoacetic acid** > 0.22 mmol/L
 - **D-Glucose-beta** > 0.14 mmol/L

Signification

- To know if patient is fasting or not

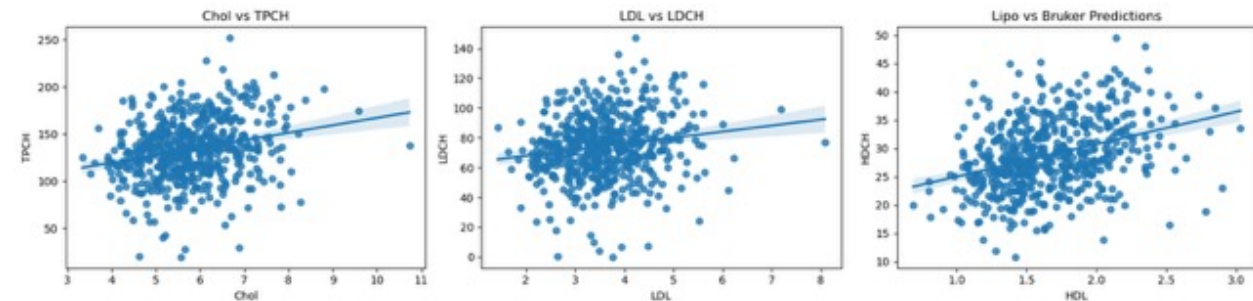
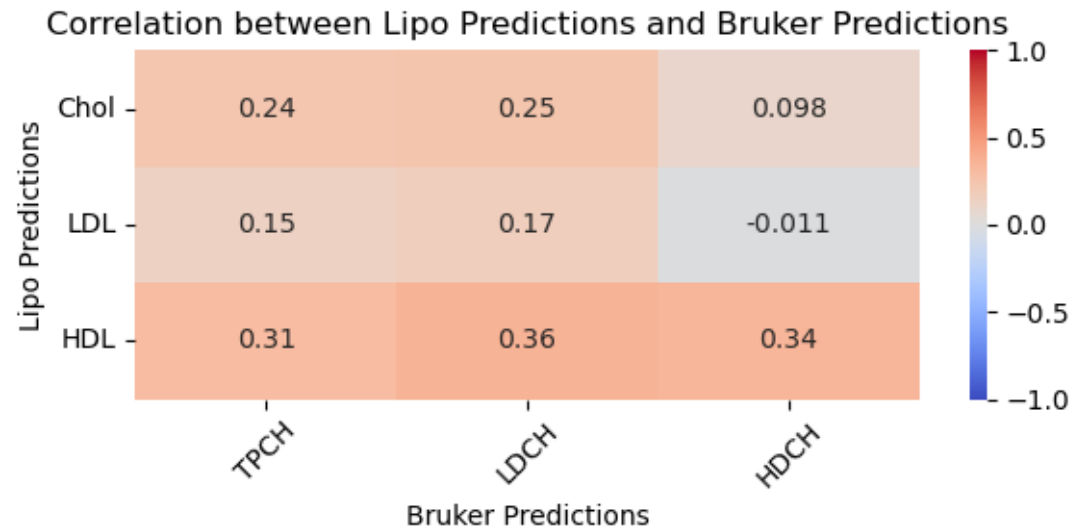


**FAILING TO FIND
BIOMARKERS IN ONE STUDY**

TENTATIVE EXPLANATION

Quality of the data

- Low levels of lipoprotein is reported relative to classical values
=> Analyze of the correlation between clinical biological parameters and NMR parameters for Cholesterol, HDL-Chol, LDL-Chol (Triglycerides)



No correlation is observed

Low lipoprotein levels



Bruker BioSpin GmbH

• IVDr report

(Example of one sample)

• Analysis Report

Bruker PhenoRisk PACS™ (Post-Acute COVID-19 Syndrome) RuO

Sample ID: 20020
Measuring Date: 21-Feb-2024 11:15:48
Reporting Date: 21-Feb-2024 12:45:01, 6 page(s)
Quantification Method: PhenoRisk PACS RuO 1.0.0

Disclaimer

RESEARCH USE ONLY: This is no clinical diagnostic analysis report. Must not be used for clinical (medical or IVD) diagnosis or for patient management! Additional concentration range information (95% range of model) provided numerically or graphically in this report must not be used for clinical diagnostic interpretation. Application of Bruker PhenoRisk PACS RuO requires the use of Bruker's B.I.Methods SOP for plasma and serum.

Results

Risk Marker	Analyte	Value	Unit (*)	95% Range of Model	Graphics
Diabetic	Glucose	3.241	mmol/L	1.730 – 6.080	
Kidney	Creatinine	0.078	mmol/L	0.060 – 0.140	
CVD	TG	54	mg/dL	53 – 490	
CVD	Chol	105	mg/dL	140 – 341	
CVD	LDL-Chol	63	mg/dL	55 – 227	
CVD	HDL-Chol	21	mg/dL	35 – 96	
CVD	LDL-Phos	38	mg/dL	37 – 121	
CVD	HDL-Phos	29	mg/dL	57 – 136	
CVD	Apo-A1	76	mg/dL	112 – 217	
CVD	Apo-B100	55	mg/dL	48 – 160	
CVD	Apo-B100/Apo-A1	0.72	-	0.30 – 1.07	
Inflammatory	GlycA	0.81	p.d.u	0.85 – 1.35	
Inflammatory	GlycB	0.31	p.d.u	0.41 – 0.68	
Inflammatory	Glyc	1.12	p.d.u	1.24 – 2.11	
Inflammatory	SPC	0.82	p.d.u	1.41 – 3.68	
Inflammatory	Glyc/SPC	1.37	-	0.41 – 1.08	

(*) Inflammation markers are reported in procedure defined units (p.d.u). Please see explanation section for details.
Yellow color indicates failed quality checks of the underlying NMR data or sample selectively for metabolites marked.
Please handle labelled results with caution.

- Abnormal low levels of lipoprotein variables (and glycoprotein variables)
- Glucose and creatinine levels seem normal

*CV is < 2% since the begin (4 years) of our activity

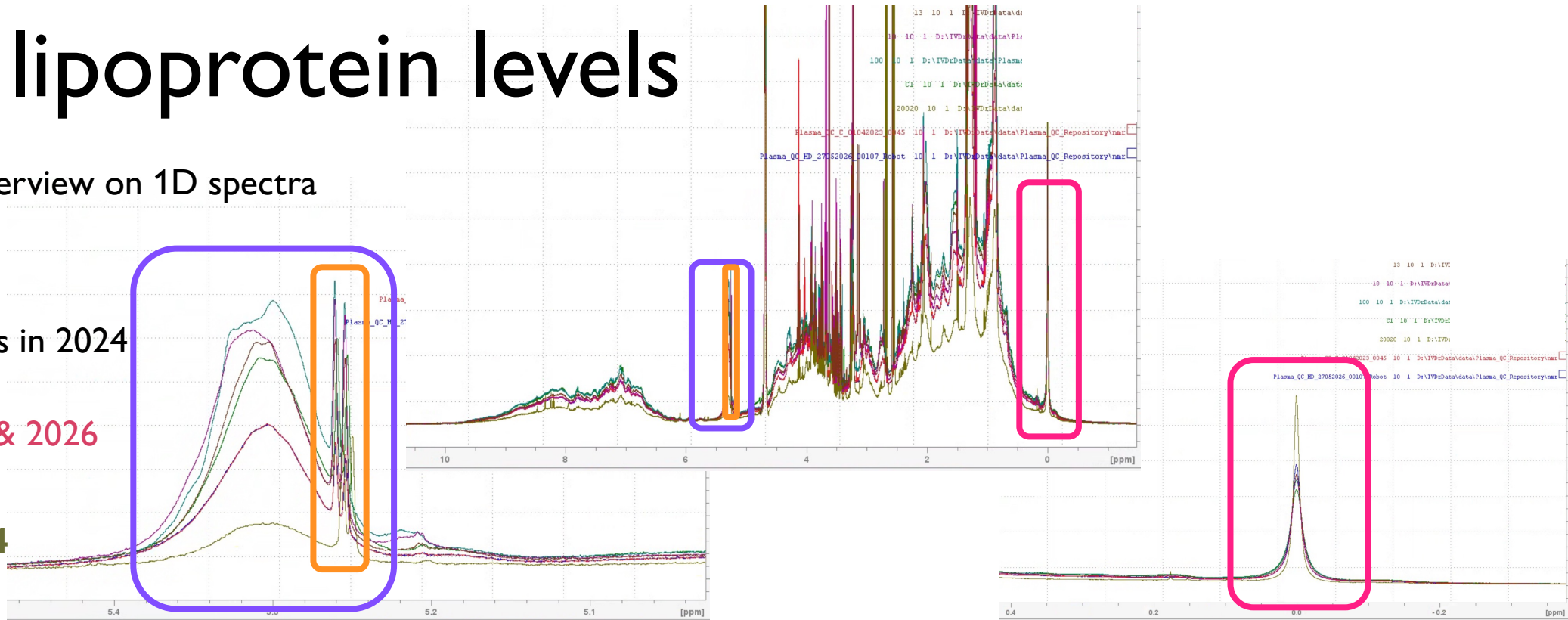
Low lipoprotein levels

Overview on 1D spectra

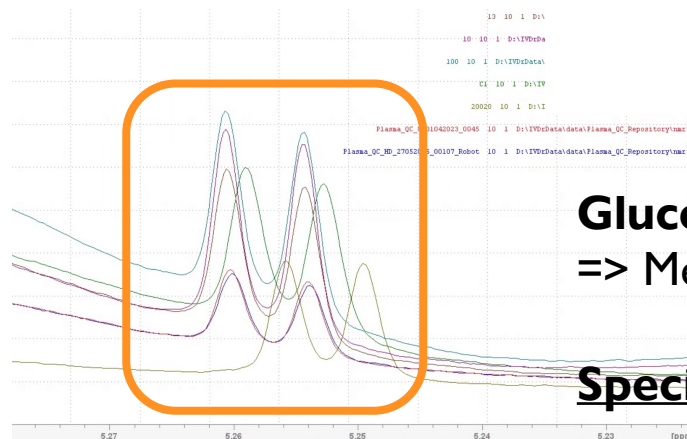
Other cohorts in 2024

QC* in 2023 & 2026

Cohort 2024



TSP signal is increased
=> Indicative of low lipoprotein levels

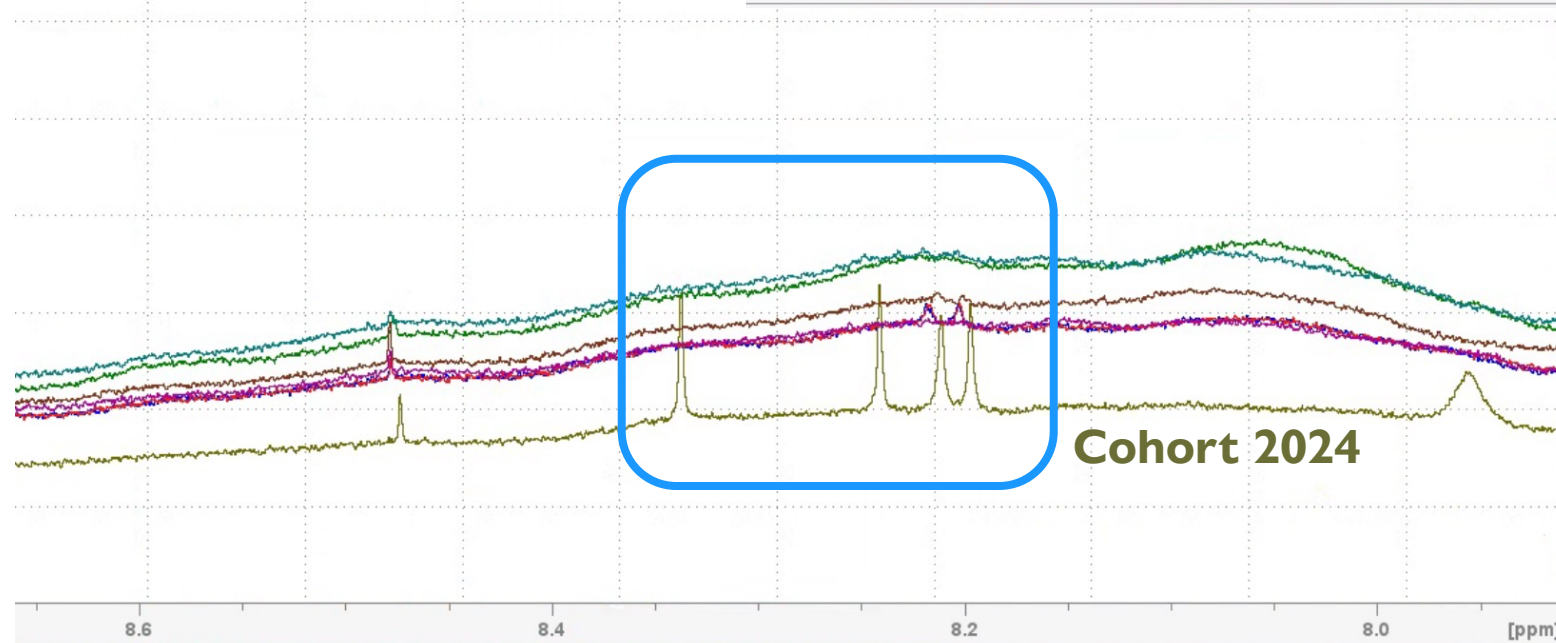
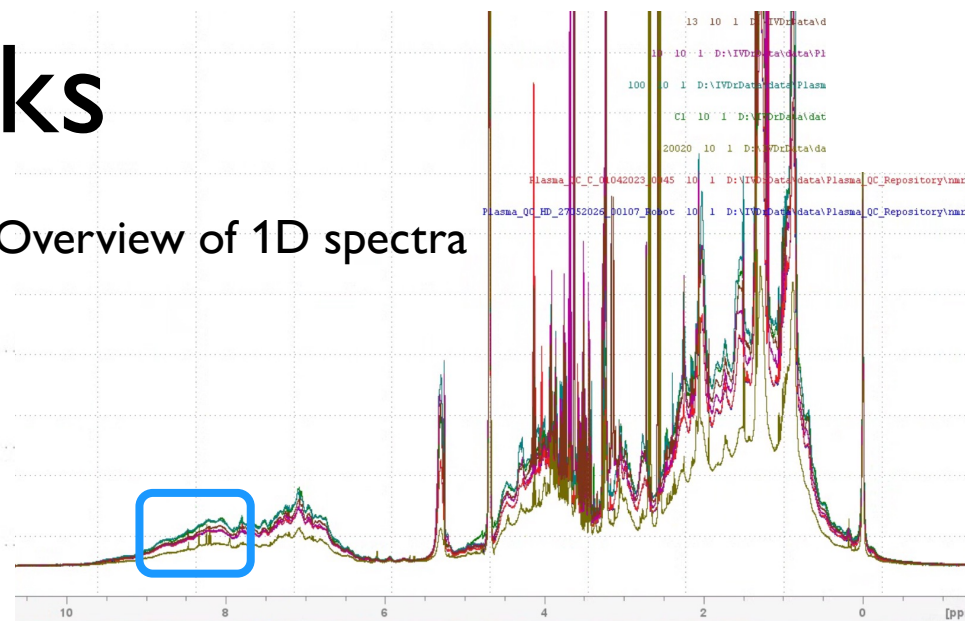


Glucose level is similar
=> Metabolite levels are not dramatically affected

Specific reduction of the lipoprotein levels

Abnormal peaks

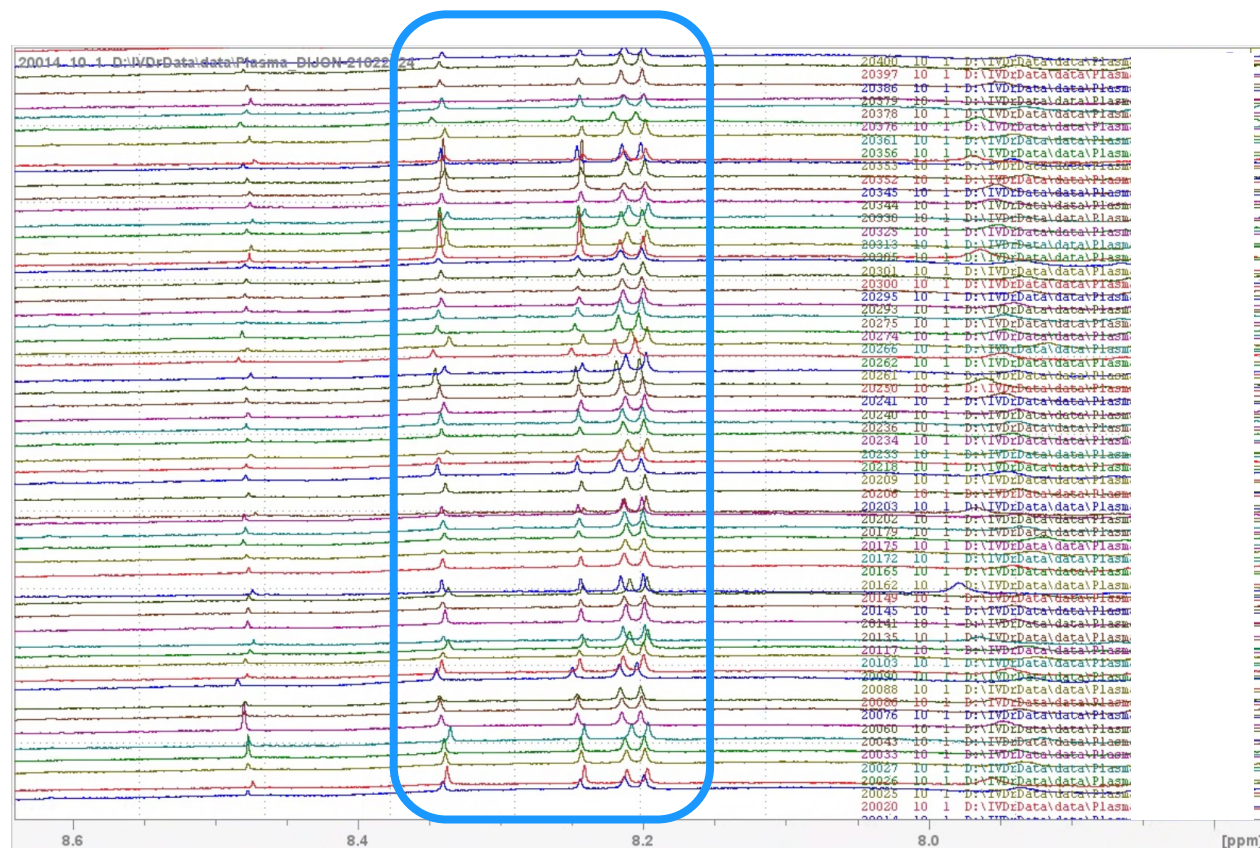
Overview of 1D spectra



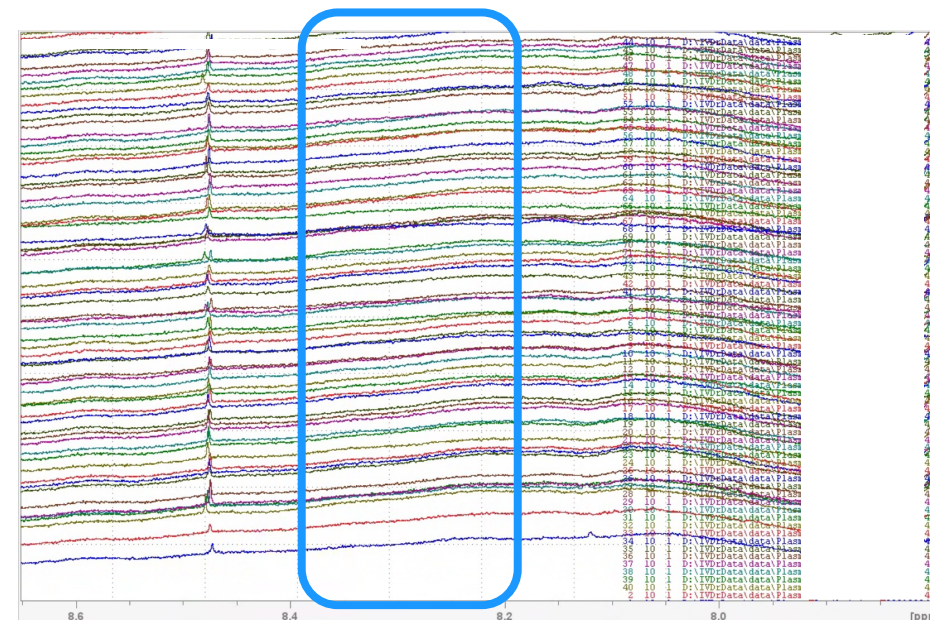
Specific peaks are observed

Abnormal peaks are specific of the cohort

Cohort 2024

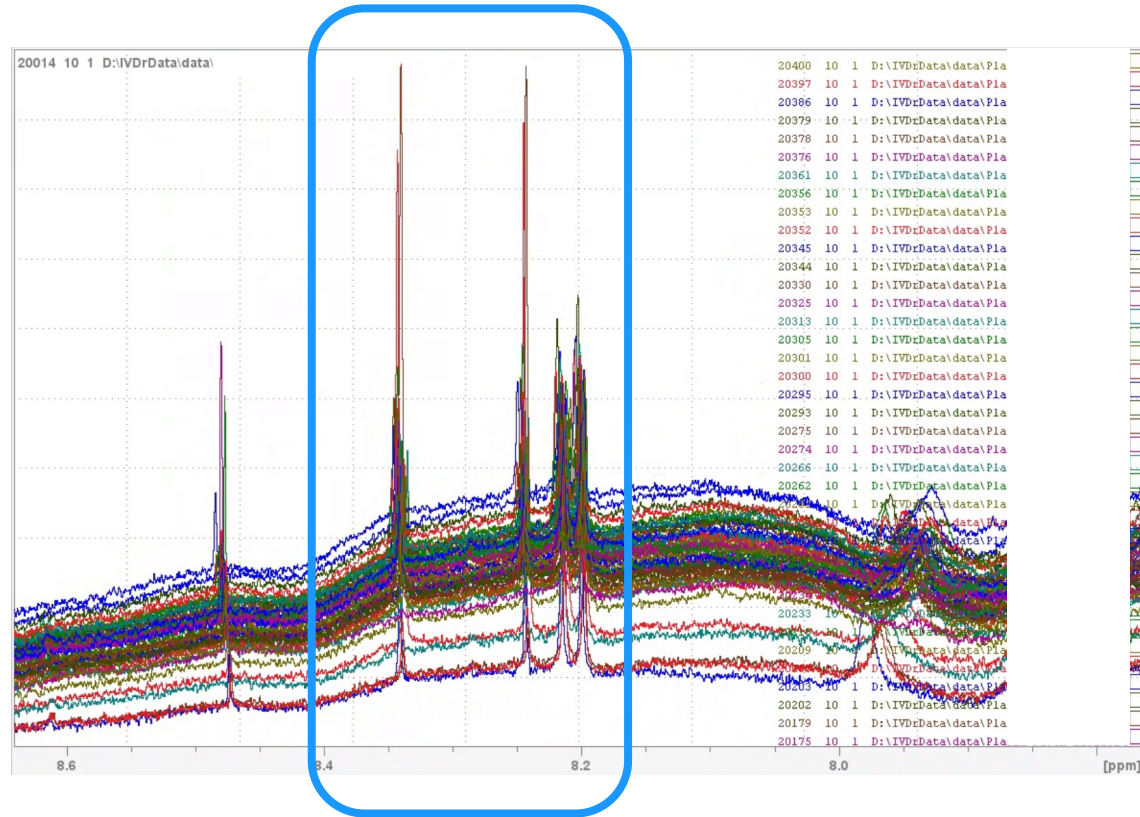


Example on another cohort

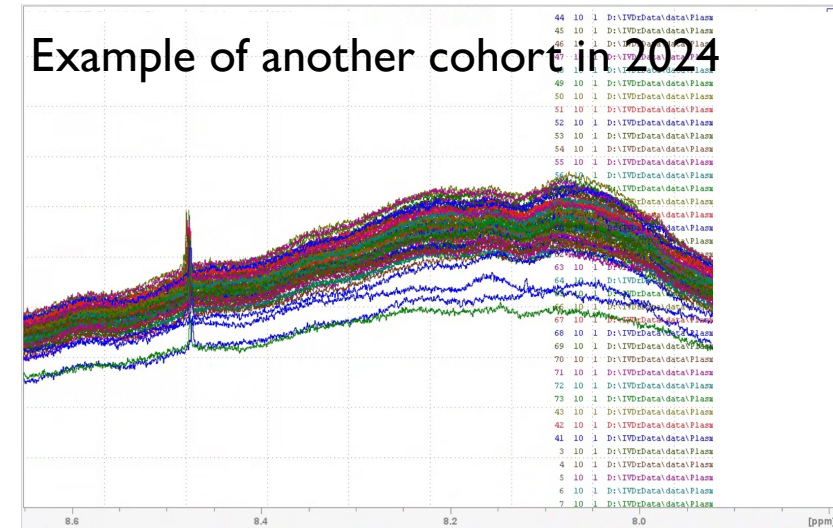


Superimposition of spectra

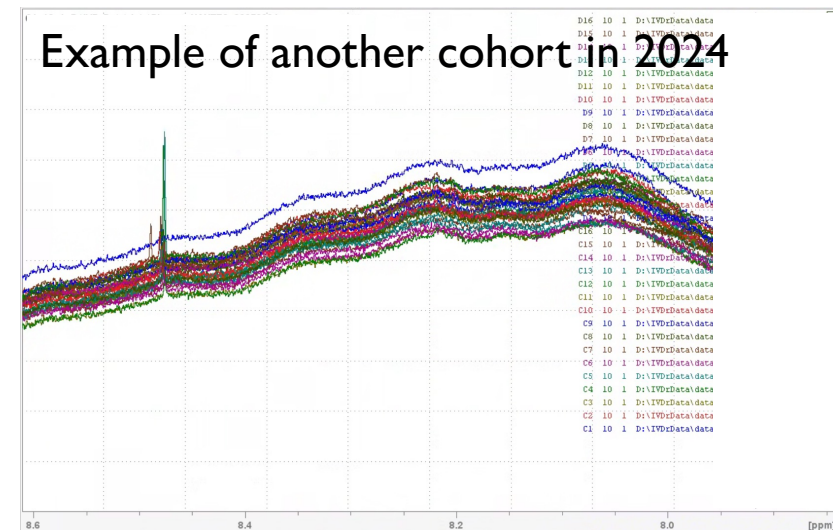
Cohort 2024



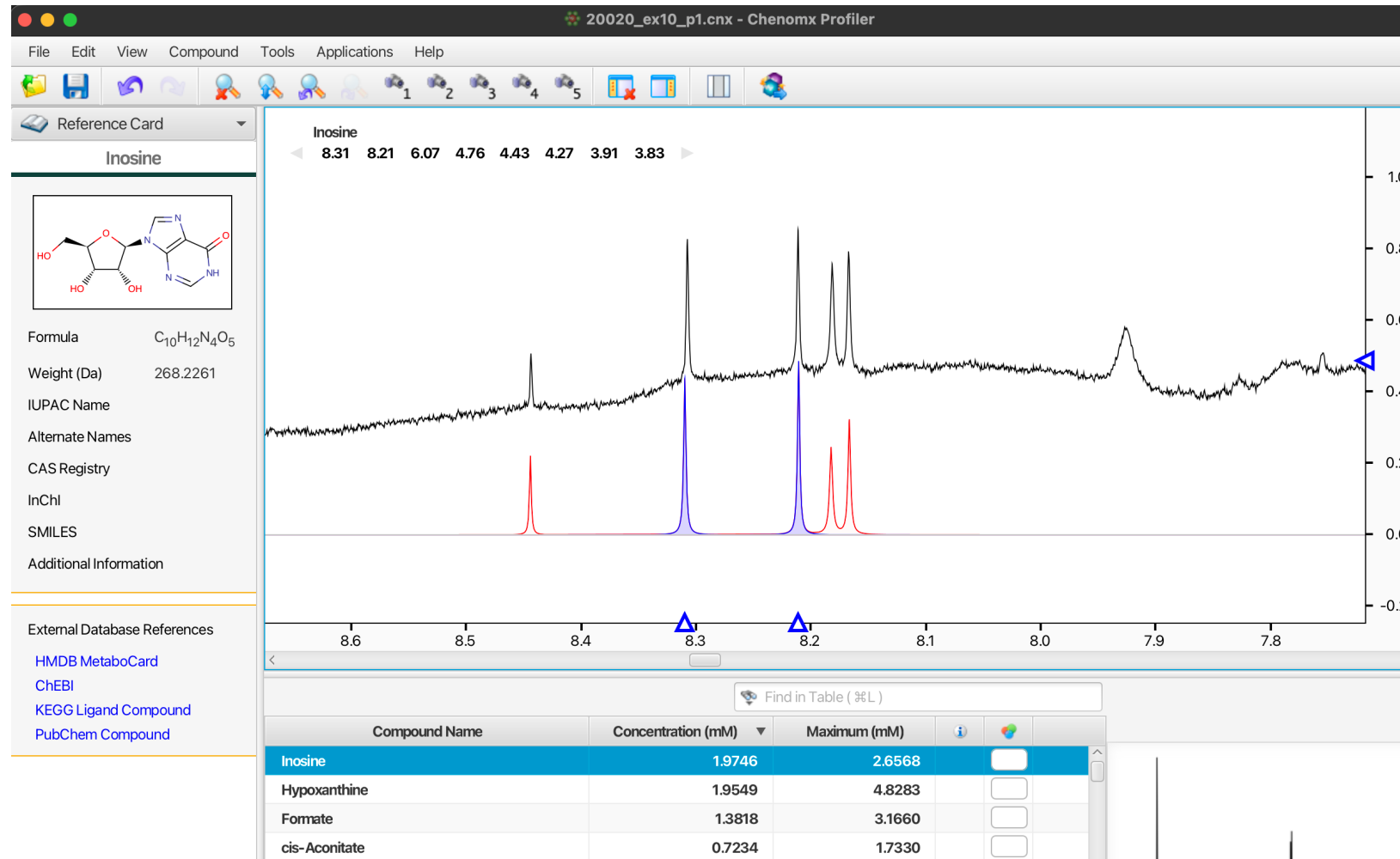
Example of another cohort in 2024



Example of another cohort in 2024



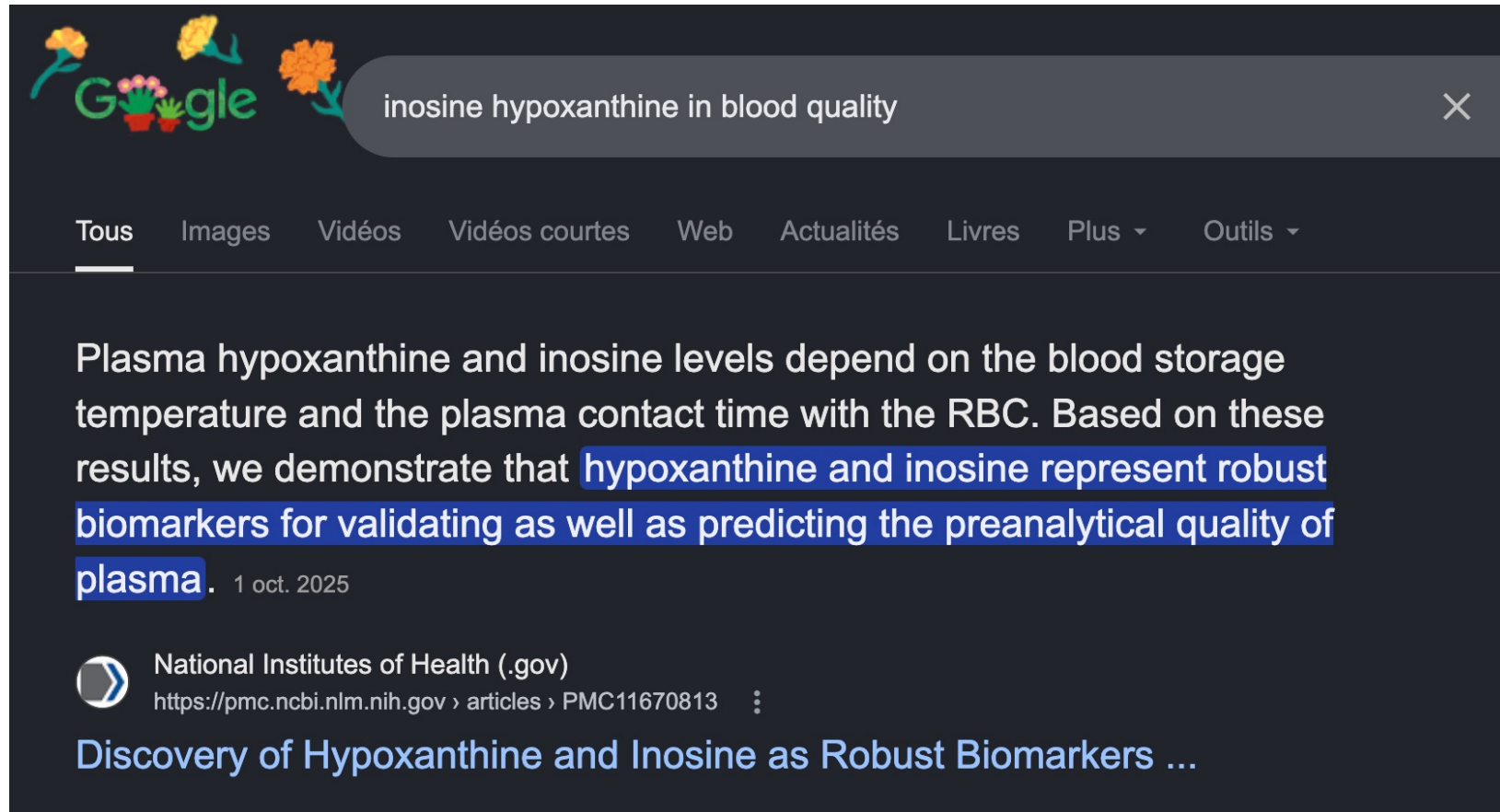
Identification of 2 abnormal compounds



✓ Inosine

✓ Hypoxanthine

What is the meaning of the presence of these two abnormal compounds ?



A screenshot of a Google search results page. The search bar at the top contains the text "inosine hypoxanthine in blood quality". Below the search bar, there are navigation tabs: "Tous", "Images", "Vidéos", "Vidéos courtes", "Web", "Actualités", "Livres", "Plus", and "Outils". The main content area displays a search result from the "National Institutes of Health (.gov)". The snippet of the article reads: "Plasma hypoxanthine and inosine levels depend on the blood storage temperature and the plasma contact time with the RBC. Based on these results, we demonstrate that hypoxanthine and inosine represent robust biomarkers for validating as well as predicting the preanalytical quality of plasma." The date "1 oct. 2025" is shown below the snippet. The source is listed as "National Institutes of Health (.gov)" with the URL "https://pmc.ncbi.nlm.nih.gov › articles › PMC11670813". The title of the article is "Discovery of Hypoxanthine and Inosine as Robust Biomarkers ...".

Google

inosine hypoxanthine in blood quality

Tous Images Vidéos Vidéos courtes Web Actualités Livres Plus Outils

Plasma hypoxanthine and inosine levels depend on the blood storage temperature and the plasma contact time with the RBC. Based on these results, we demonstrate that hypoxanthine and inosine represent robust biomarkers for validating as well as predicting the preanalytical quality of plasma. 1 oct. 2025

National Institutes of Health (.gov)
https://pmc.ncbi.nlm.nih.gov › articles › PMC11670813

Discovery of Hypoxanthine and Inosine as Robust Biomarkers ...

What is the meaning of these compounds?

► [Anal Chem](#). Author manuscript; available in PMC: 2025 Oct 1.

Published in final edited form as: *Anal Chem*. 2024 Sep 18;96(39):15754–15764. doi:

[10.1021/acs.analchem.4c03719](https://doi.org/10.1021/acs.analchem.4c03719) 

Article published in **Oct 2025**

Discovery of Hypoxanthine and Inosine as Robust Biomarkers for Predicting the Preanalytical Quality of Human Plasma and Serum for Metabolomics

[G A Nagana Gowda](#)^{1,2,*}, [Vadim Pascua](#)¹, [Lucas Hill](#)¹, [Danijel Djukovic](#)^{1,2}, [Dennis Wang](#)³, [Daniel Raftery](#)^{1,2,4,*}

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PMCID: PMC11670813 NIHMSID: NIHMS2041377 PMID: [39291745](#)

The publisher's version of this article is available at [Anal Chem](#) 

Abstract

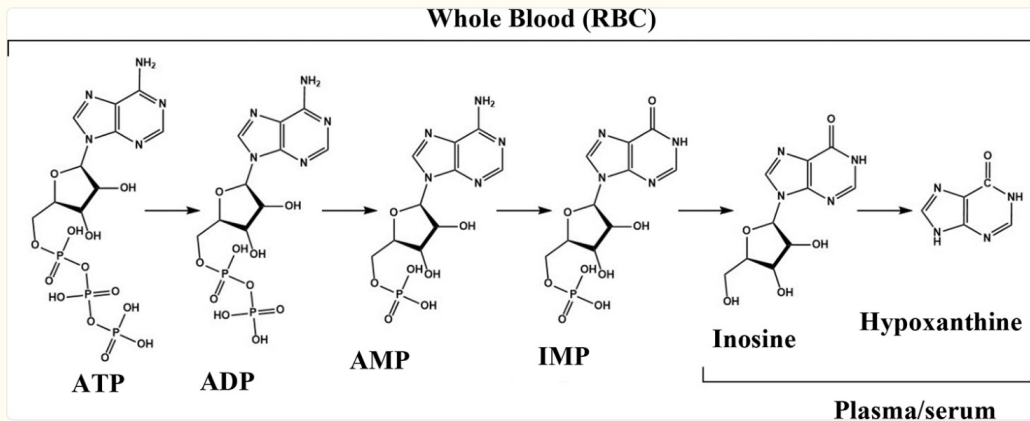
In cold human blood, the anomalous dynamics of adenosine triphosphate (ATP) result in the progressive accumulation of adenosine diphosphate (ADP), adenosine monophosphate (AMP), inosine monophosphate (IMP), inosine, and hypoxanthine. While the ATP, ADP, AMP, and IMP are confined to red blood cells (RBCs), inosine and hypoxanthine are excreted into plasma/serum. The plasma/serum levels of inosine and hypoxanthine depend on the temperature of blood and the plasma/serum contact time with the RBCs and hence they represent robust biomarkers for evaluating the preanalytical quality of plasma/serum. These biomarkers are highly specific since they are generally not present or at very low levels in fresh plasma/serum and are highly sensitive since they are derived from ATP, one of the most abundant metabolites in blood. Further, whether or not blood was kept at room temperature or on ice could be predicted based on inosine levels. An analysis of >2000 plasma/serum samples processed for metabolomics-centric analyses showed alarmingly high levels of inosine and hypoxanthine. The results highlight the gravity of sample quality challenges with high risk of grossly inaccurate measurements and incorrect study outcomes. The discovery of these robust biomarkers provides new ways to address the longstanding and underappreciated preanalytical sample quality challenges in the blood metabolomics field.

Keywords: Human blood, plasma, serum, red blood cells (RBCs), ¹H NMR spectroscopy, anomalous dynamics, adenosine triphosphate, adenosine diphosphate, adenosine monophosphate, inosine monophosphate, inosine, hypoxanthine, preanalytical sample quality

What is the meaning of these compounds?

Origin

Scheme 1:

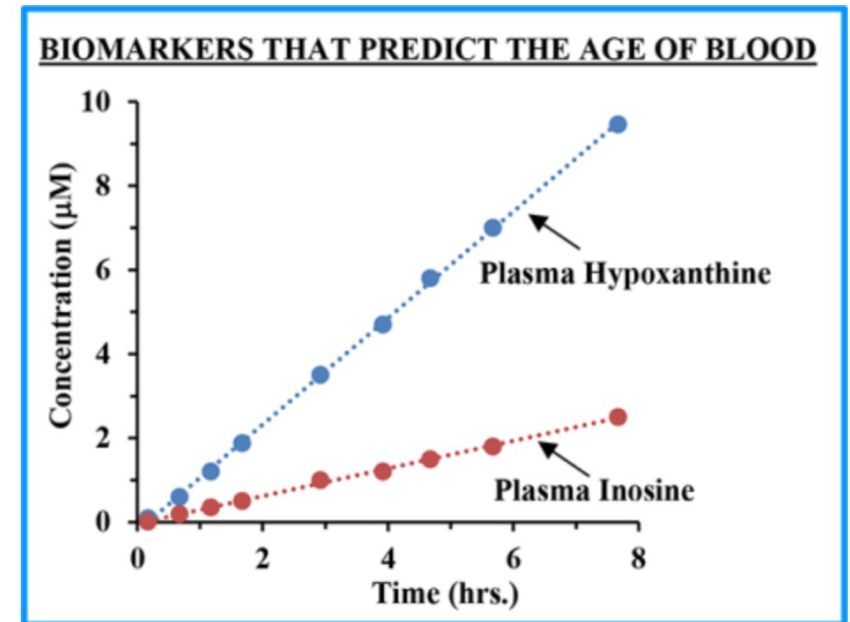


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In cold human blood, the anomalous dynamics of ATP in red blood cell (RBC) leads to the progressive accumulation of adenosine diphosphate (ADP), adenosine monophosphate (AMP), inosine monophosphate (IMP), inosine, and hypoxanthine. The phosphorous containing metabolites, ATP, ADP, AMP, and IMP, are confined to the RBC due to their negative charge and high polarity, whereas the non-phosphorus metabolites, inosine and hypoxanthine, are excreted into the plasma/serum.

Age of blood

Graphical Abstract

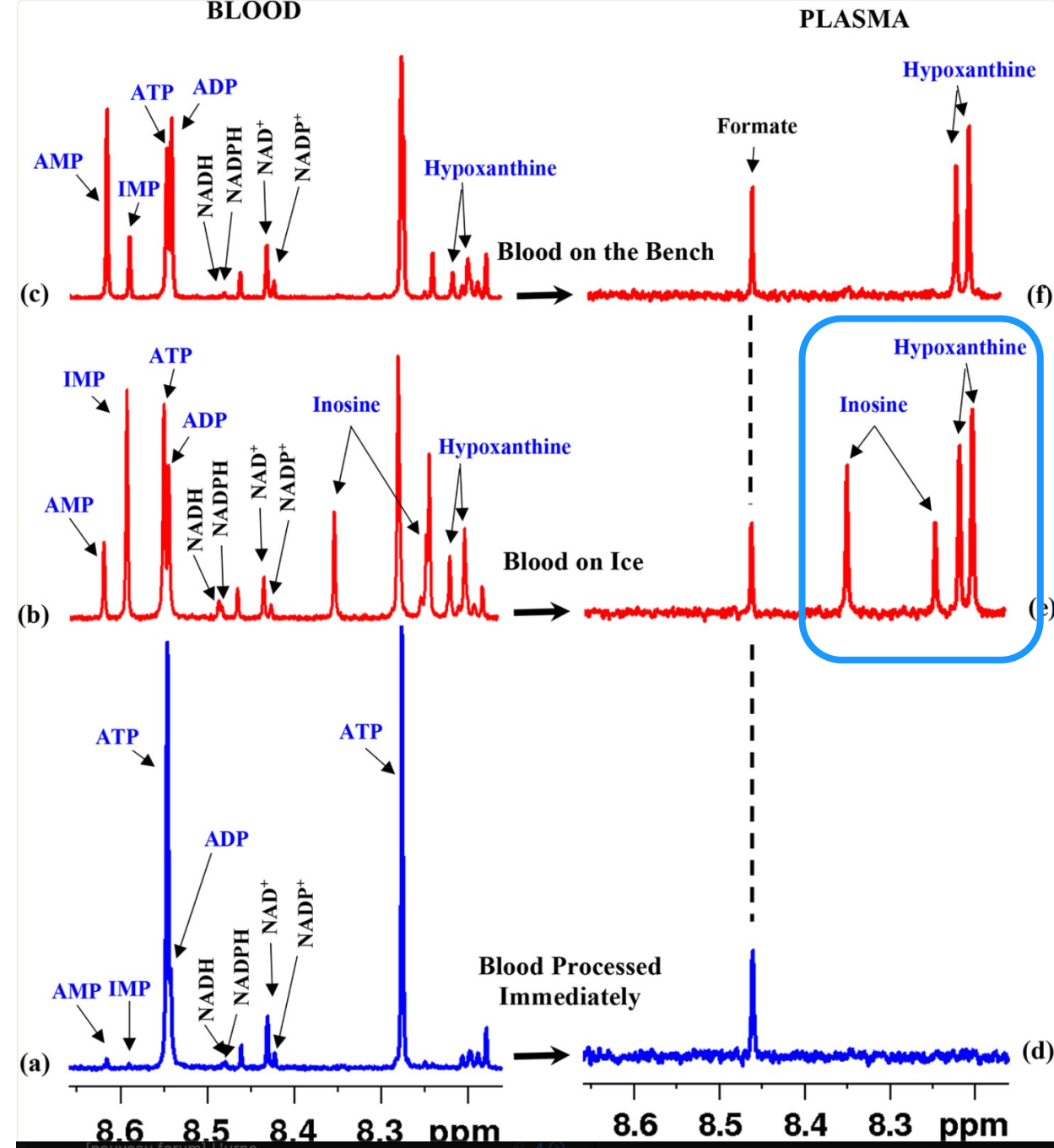


Blood preparation

⇒ Related to **conditions of plasma collection**

⇒ The presence of Inosine indicate that **blood was kept on Ice**

⇒ **Immediately separate RBC** from plasma should be done



Blood preparation

=> Delay of preparation is crucial for good quality of plasma

In our cohort, this is even worse situation...
Inosine level is similar to Hypoxanthine

=> Probably kept on Ice, about 24 hours

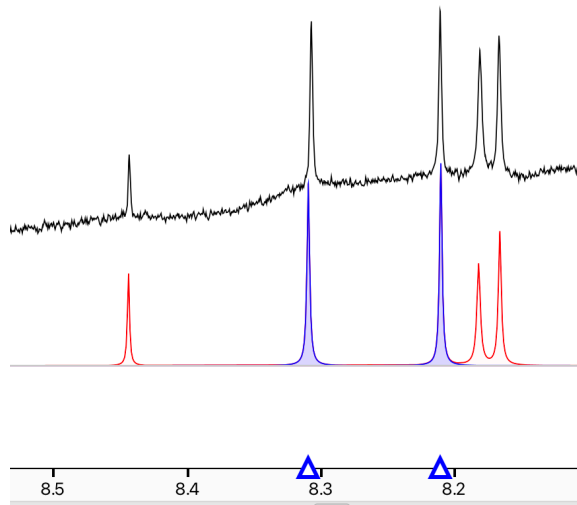
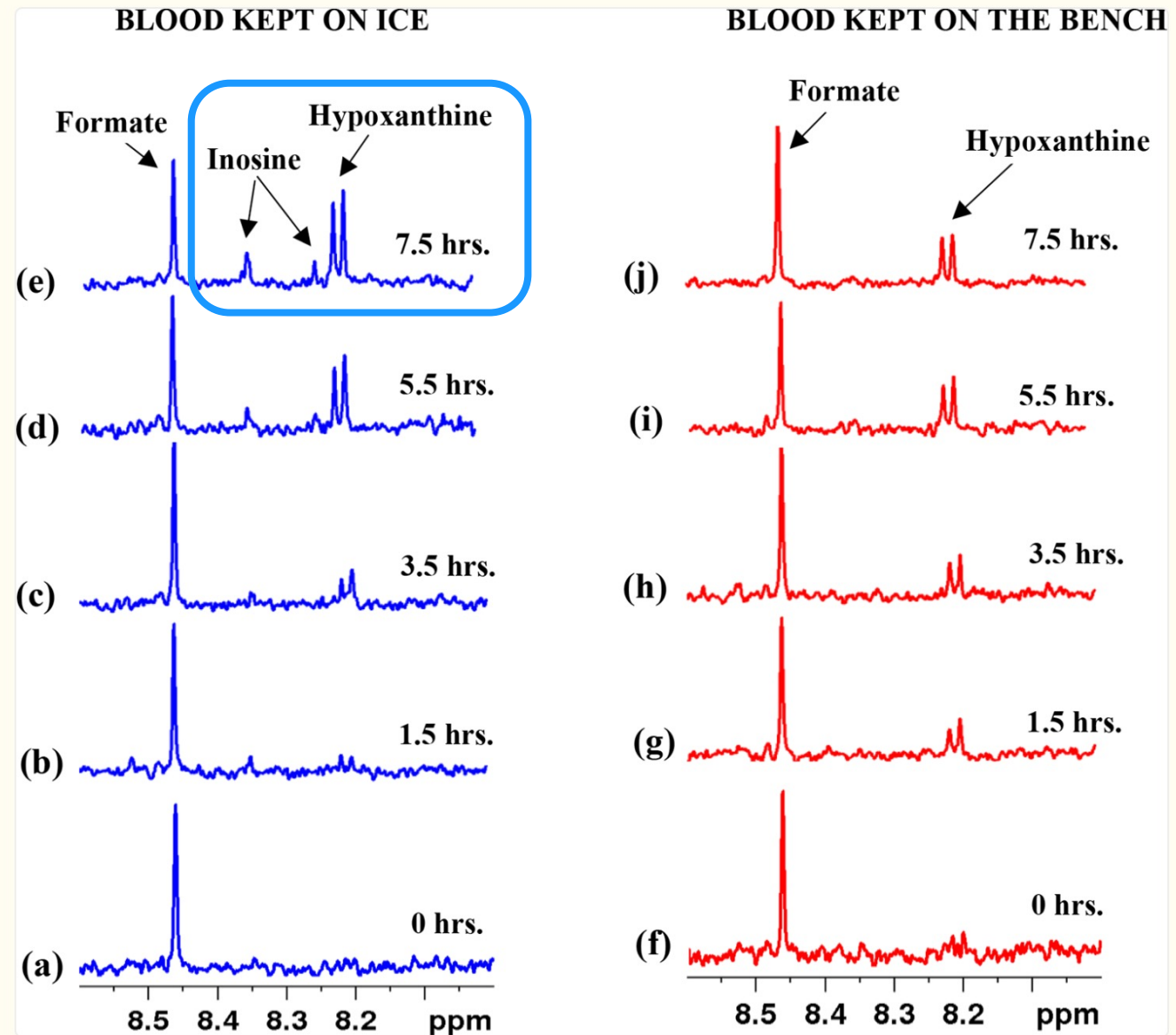


Figure 3:



Plasma

Subject 10

ICE

BENCH AT RT

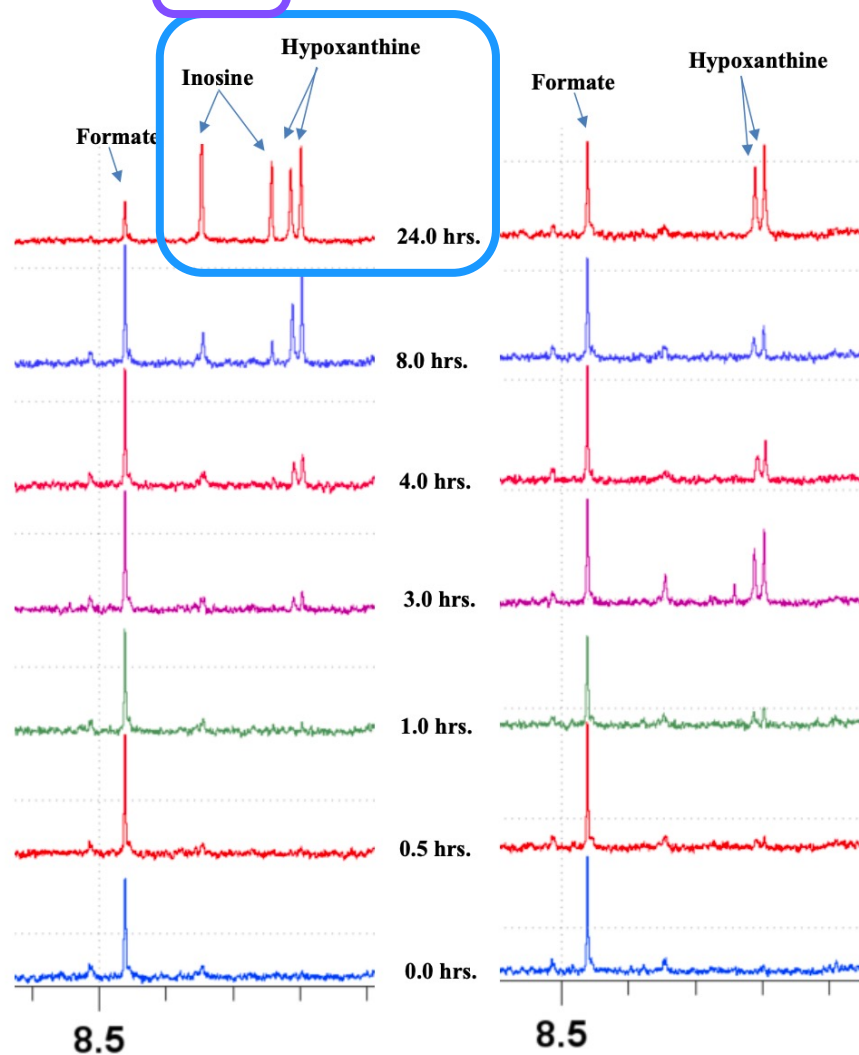


Figure S15. Portions of 800 MHz ^1H NMR spectra of plasma from the same subject (subject 10). Plasma was separated from blood placed on ice or the bench at room temperature (RT) for different times as indicated.

Serum

SERUM FROM BLOOD KEPT ON THE BENCH AT RT

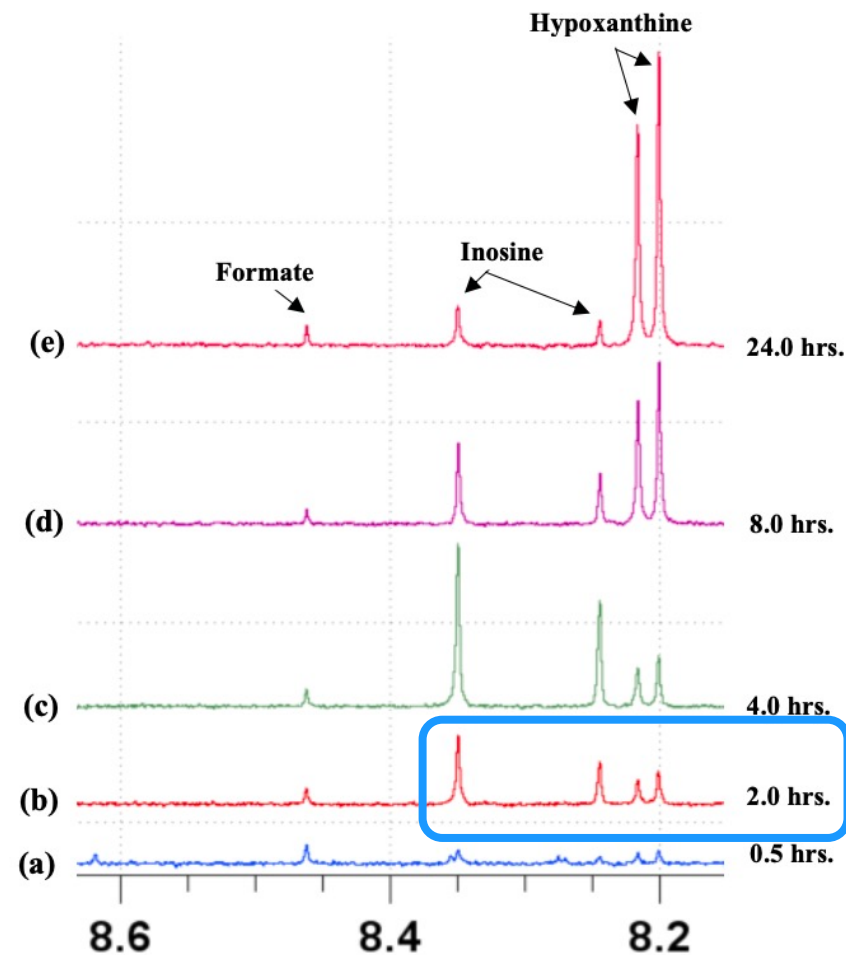
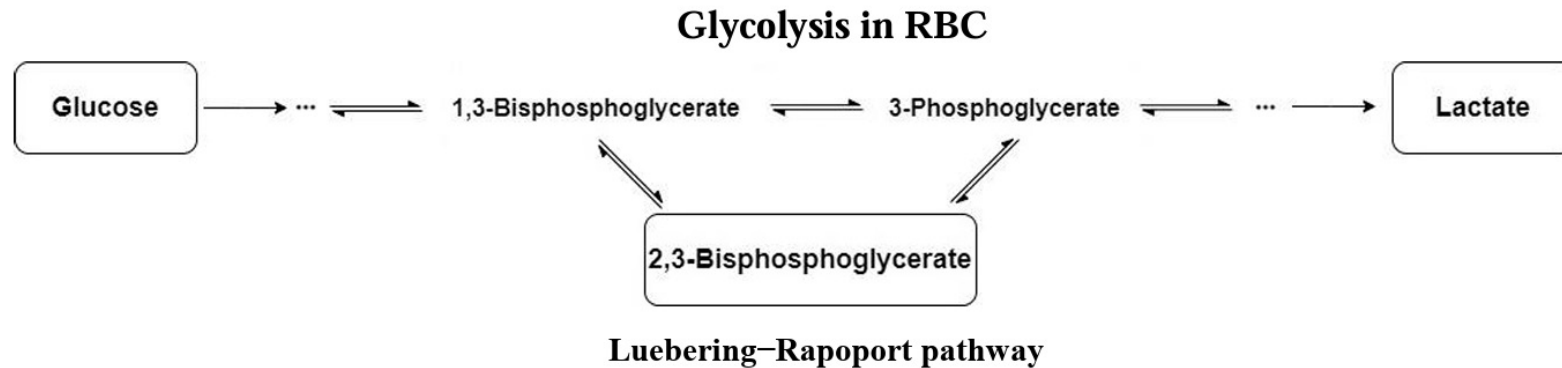


Figure S16. Portions of 800 MHz ^1H NMR spectra of serum from the same subject. Serum was separated from blood placed on the bench, at room temperature (RT), for 0.5, 2.0, 4.0, 8.0, or 24.0 hrs.

Is it possible to use only metabolite reports?

Unfortunately, the long-term presence of RBC prior to separation of plasma **also affects the level of important metabolites** such Glucose, Lactate, ...



Scheme S1. Glycolysis pathway in RBC. In addition to ATP, glucose, 2,3-bisphosphoglycerate, and lactate are the three most abundant metabolites in blood.

Conclusion:

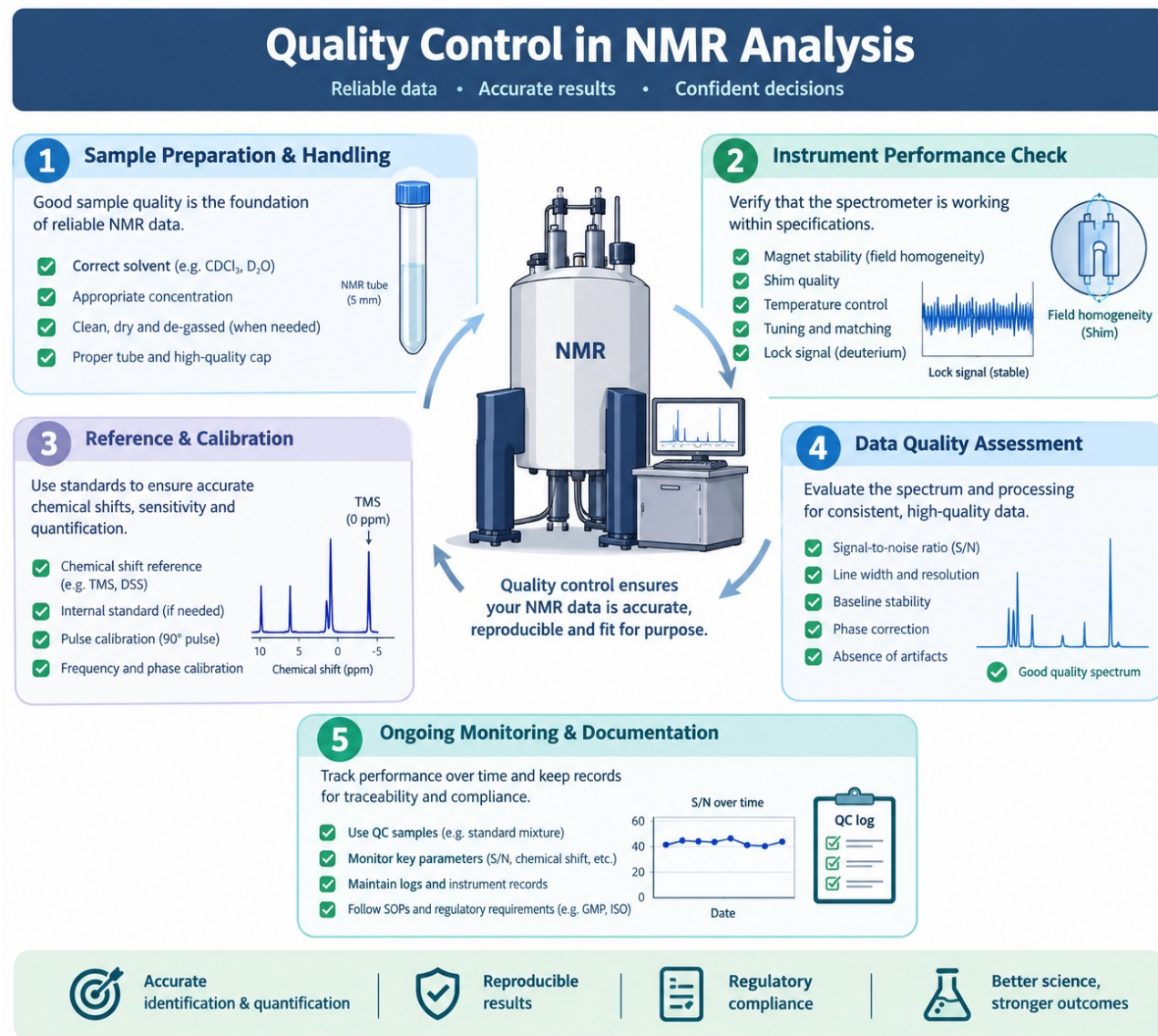
Interest of NMR for the evaluation of biobank quality

In conclusion, we describe the **discovery of hypoxanthine and inosine as potential biomarkers to assess or validate the preanalytical quality of plasma/serum**, a widely used biospecimen in the metabolomics field. These biomarkers are highly specific since they are generally not present in plasma/serum and they accumulate in blood as it ages either at room temperature or even at cold temperatures due to the anomalous dynamics of ATP in cold blood. They are highly sensitive since they are derived from ATP, which is one of the highly concentrated metabolites in human blood. Interestingly, our results also show that, **based on plasma inosine levels, whether or not blood was kept at room temperature or on ice could be predicted**. To the best of our knowledge these are the only biomarkers associated with the metabolism of blood cells that are excreted into plasma at high concentrations enough to detect early even by NMR spectroscopy. Such characteristics enable their detection routinely by both NMR and MS. The fact that these biomarkers are derived from a highly unstable and highly concentrated blood metabolite, ATP, gives support for their **high sensitivity and specificity**. Their surprisingly high levels found in sample cohorts procured based on metabolomics-centric protocols indicate the gravity of the compliance challenges for sample procurement. Henceforth, in addition to their concentrations, the characteristic region (8.2 to 8.4 ppm) of ^1H NMR spectra of plasma/serum that show hypoxanthine and inosine peaks should be used as an important visual yardstick to evaluate the quality of plasma/serum samples used in metabolomics studies on a routine basis. Both NMR and MS are widely used in blood metabolomics. While MS is highly sensitive, NMR is highly quantitative and reproducible. Hence, **NMR is more suitable for discovering biomarkers and developing methods for plasma/serum preanalytical quality assessment**. Since MS invariably uses protein precipitation prior to analyses, biomarkers developed using NMR, under similar conditions, can be directly translated for MS analyses. For this reason, in this study, we used NMR to develop and describe the biomarkers' discovery and then demonstrated their utility even in MS analysis. Overall, the discovery of robust biomarkers provides new opportunities to **address the grossly underrecognized and underappreciated challenge of sample quality in the metabolomics field**.

Quality Control in NMR analysis

- ✓ **Robust**
- ✓ **Quantitative**

**Great advantage of
NMR for biobank
control quality**

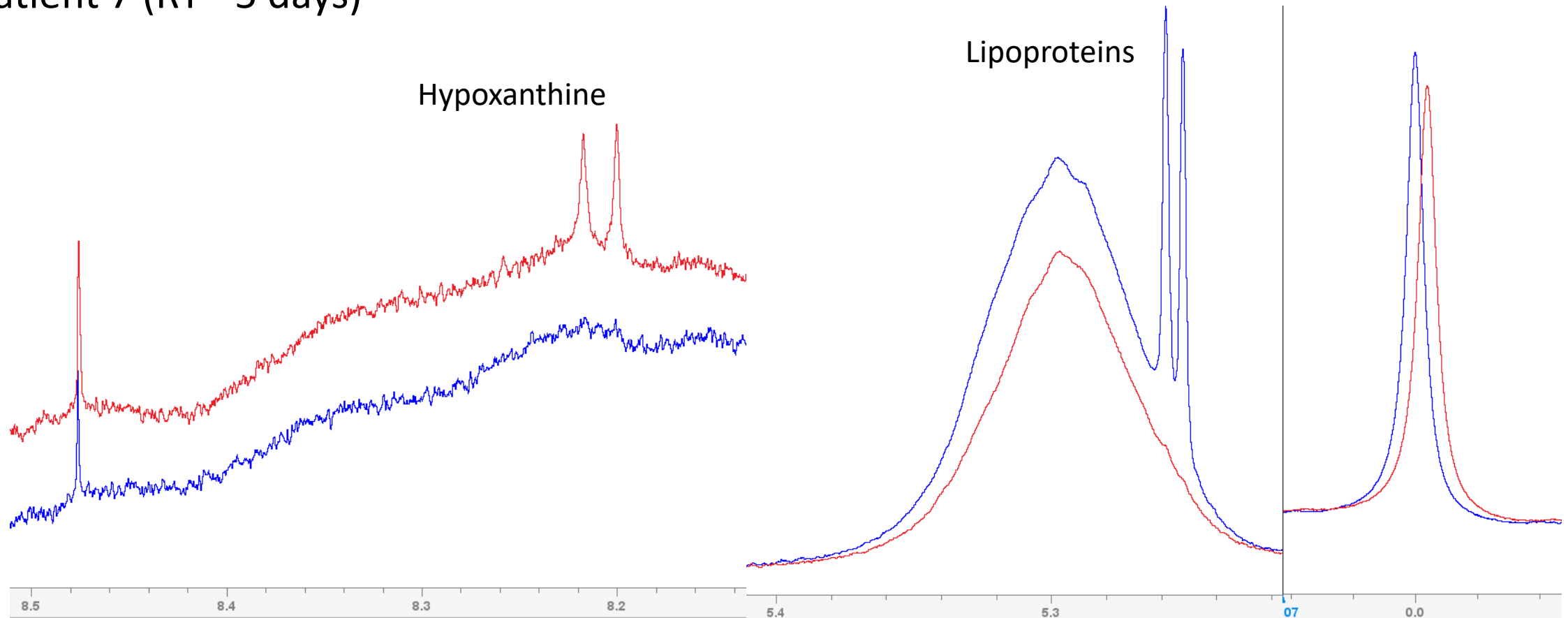


Example of evolution at RT or 4°C before centrifugation

- Is it possible to correct the evolution of sample ?

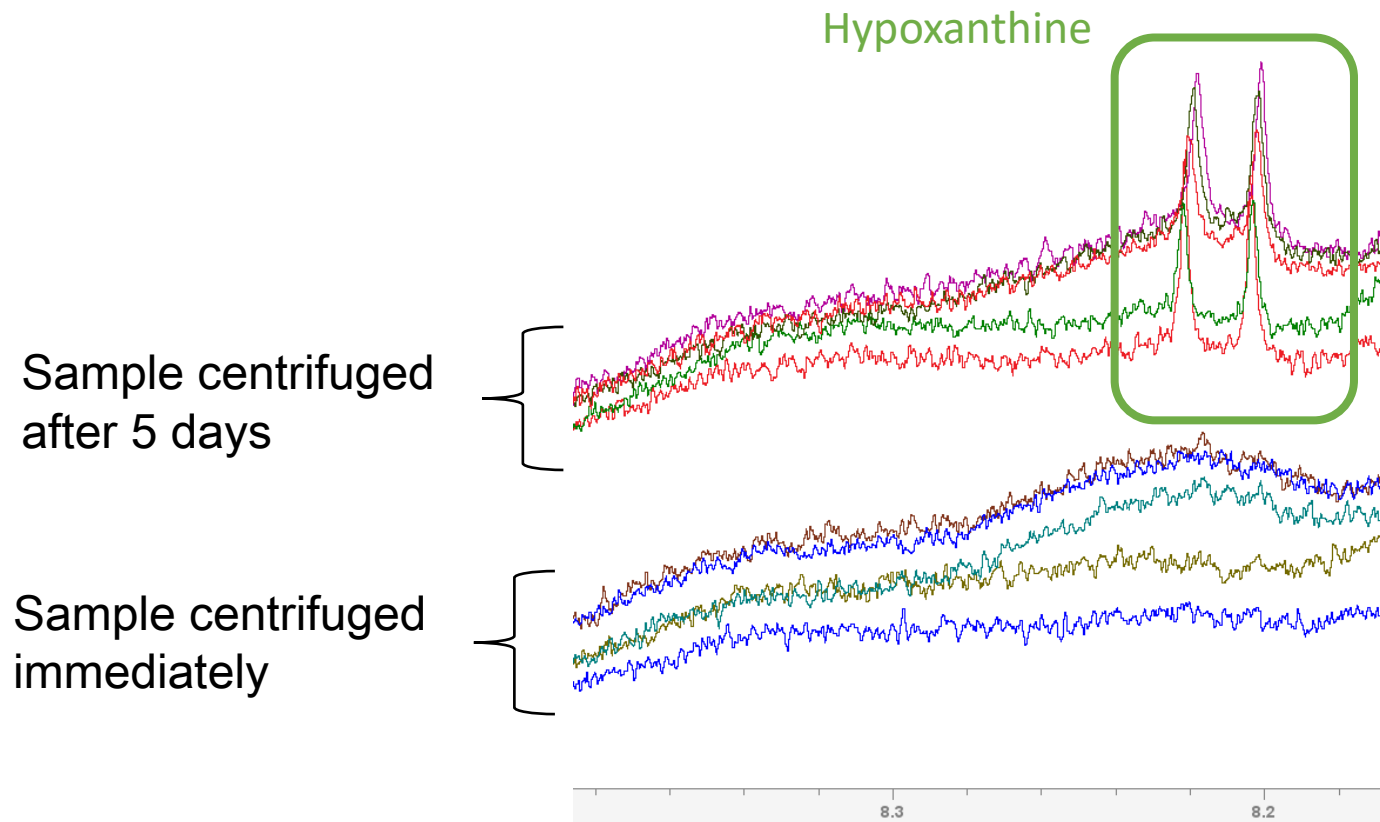
Example of evolution

Patient 7 (RT - 5 days)



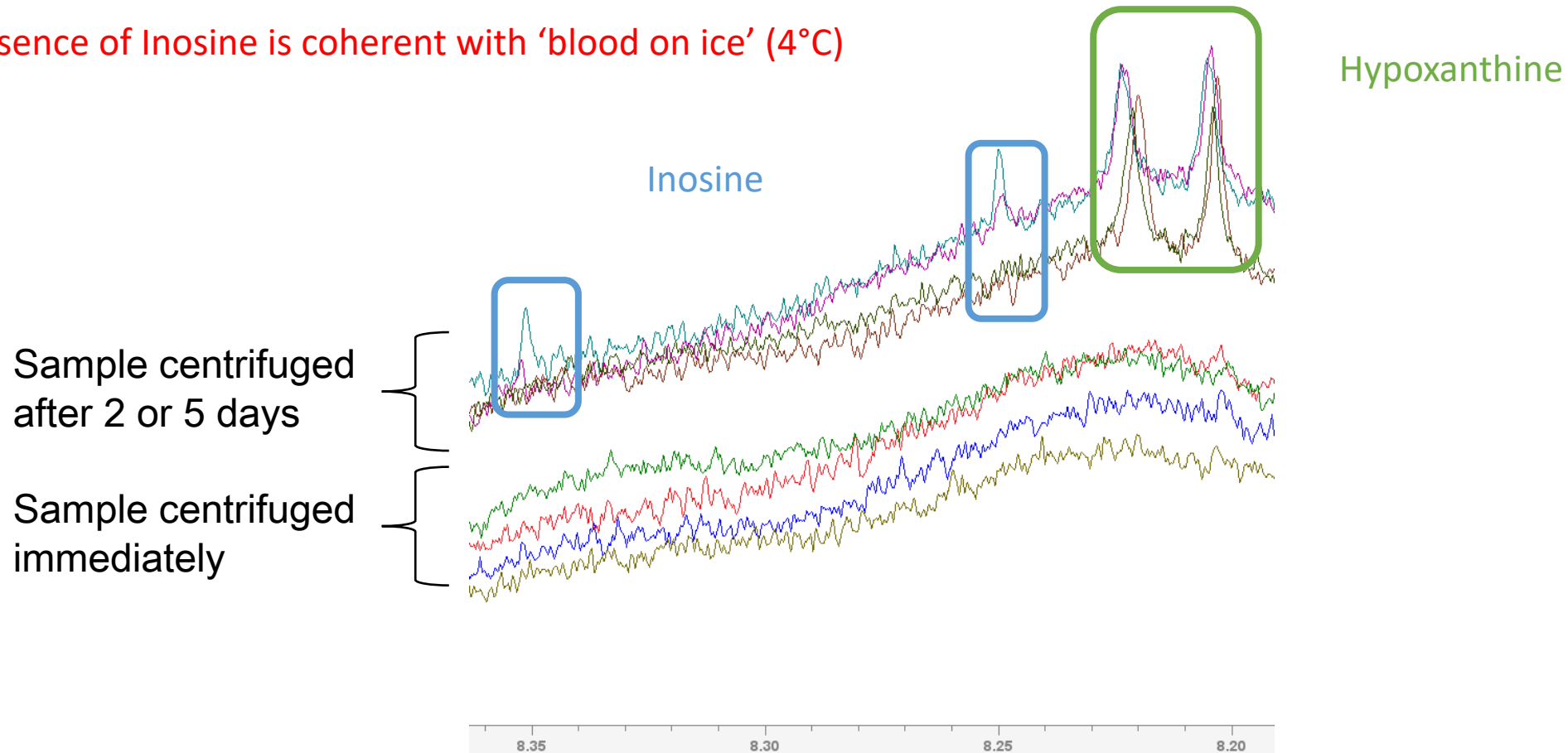
Focus Patients 3-8 – hypoxanthine/inosine

Absence of Inosine is coherent with 'blood on the bench' (RT)



Focus Patients A-D – hypoxanthine/inosine

Presence of Inosine is coherent with 'blood on ice' (4°C)



Variations in %

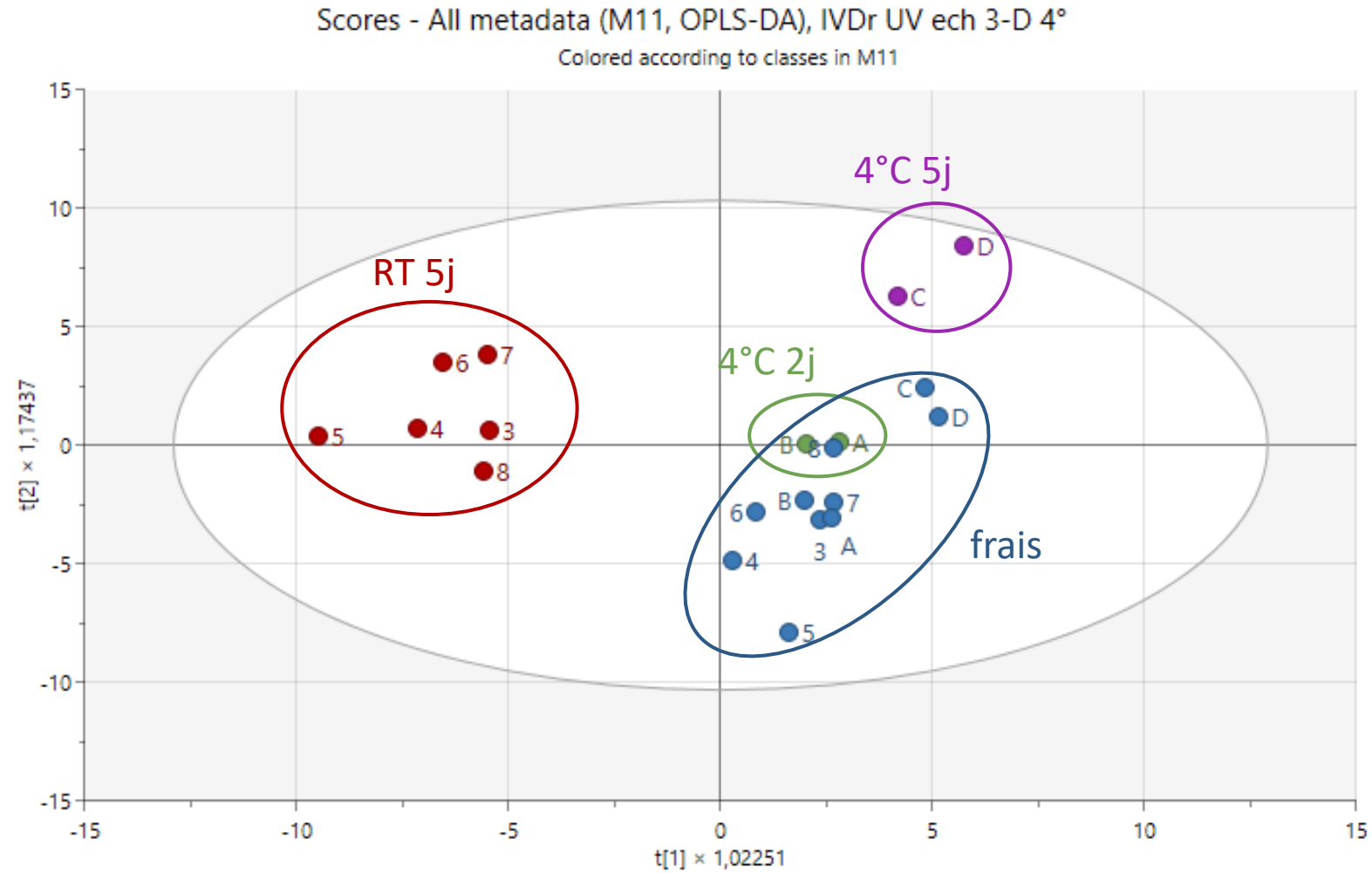
(after days - immediate)/immediate centrifugation

Patient	TPTG [mg/dL]	TPCH [mg/dL]	LDCH [mg/dL]	HDCH [mg/dL]	TBPN [nmol/L]	VLTG [mg/dL]	IDTG [mg/dL]	LDTG [mg/dL]	HDTG [mg/dL]	
2	-69%	-11%	-16%	19%	-37%	-84%	-100%	-48%	-45%	RT
3	6%	11%	7%	4%	11%	-8%	2%	16%	7%	
4	-14%	0%	-4%	17%	-1%	-6%	-100%	-12%	-42%	
6	-28%	-4%	-10%	4%	0%	-44%	-93%	12%	-35%	
7	-57%	-27%	-21%	-31%	-17%	-18%	-100%	-30%	-47%	
8	3%	57%	49%	68%	43%	-4%	#DIV/0!	2%	-61%	
A	-9%	-5%	-18%	0%	-14%	-38%	-5%	-15%	-11%	
B	-7%	0%	-2%	0%	-1%	-13%	1%	-19%	-13%	
C	-4%	-6%	-15%	-2%	-10%	-14%	-15%	-13%	-8%	4°C
D	-16%	-13%	-23%	-14%	-14%	-20%	-20%	-15%	-10%	

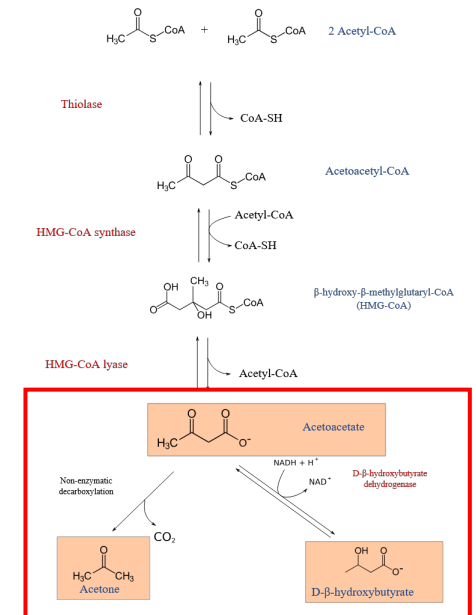
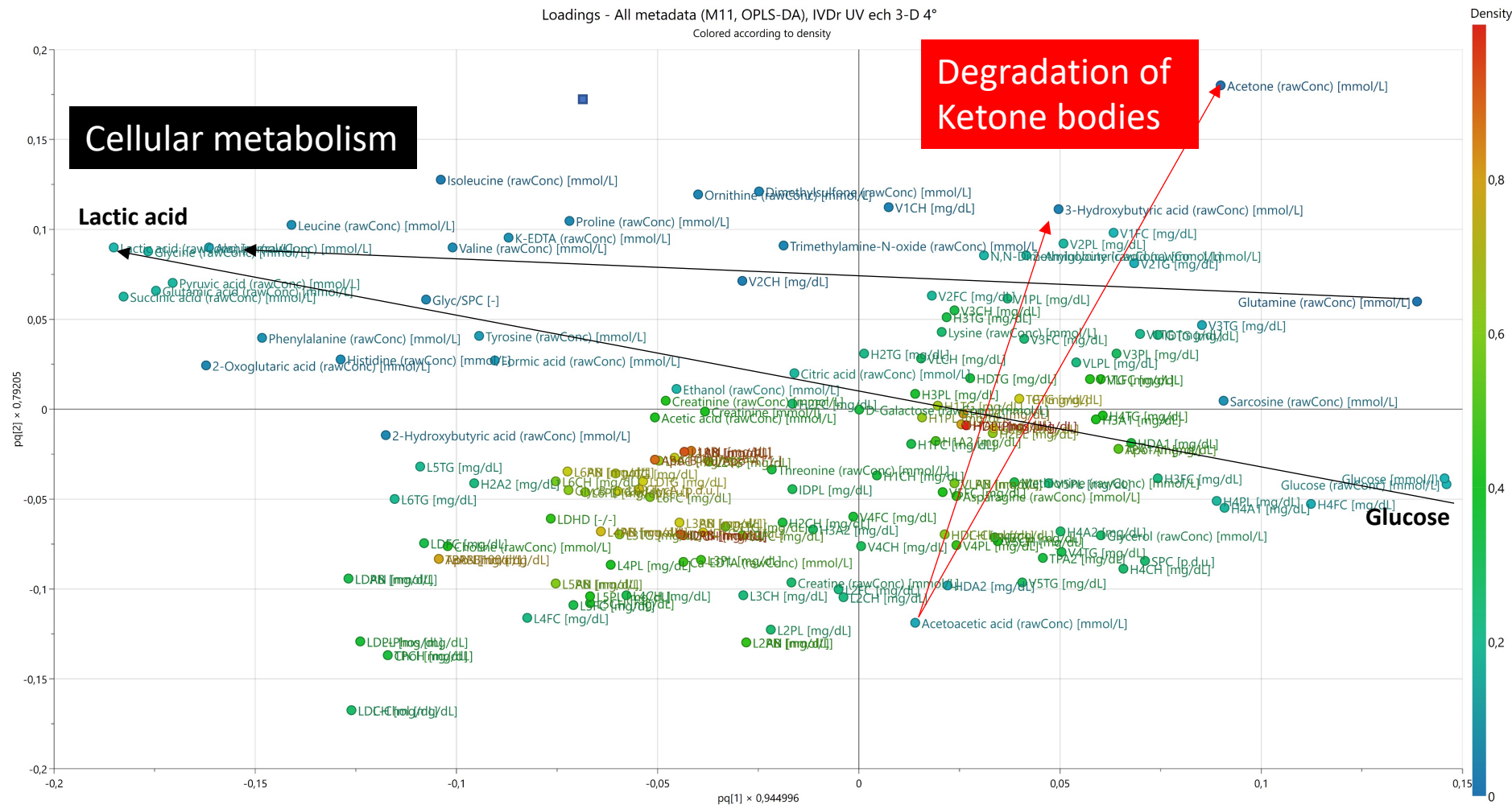
- Different evolutions of parameters on same conditions
 - Patient-dependent evolution of samples
- Therefore, it is not possible to correct the results according to time-evolution or temperature

Patient	Glutamic acid [mmol/L]	Glutamine [mmol/L]	Glycine [mmol/L]	Histidine [mmol/L]	Isoleucine [mmol/L]	Leucine [mmol/L]	Methionine [mmol/L]	Acetic acid [mmol/L]	Citric acid [mmol/L]	Formic acid [mmol/L]	Lactic acid [mmol/L]	Succinic acid [mmol/L]	Pyruvic acid [mmol/L]	Glucose [mmol/L]	
2	154%	-72%	78%	12%	-8%	23%	-81%	-22%	9%	-80%	473%	26%	329%	-80%	RT
3	125%	-48%	64%	9%	2%	64%	-24%	9%	6%	149%	581%	540%	375%	-82%	
4	69%	-100%	172%	58%	88%	100%	167%	489%	76%	108%	737%	2333%	263%	-98%	
6	317%	-29%	111%	187%	37%	117%	-62%	-46%	-24%	-8%	646%	2300%	232%	-97%	
7	190%	-49%	102%	75%	59%	162%	-70%	-76%	2%	71%	673%	775%	818%	-96%	
8	243%	-22%	138%	33%	56%	57%	-54%	-83%	-45%	17%	923%	169%	469%	-100%	
A	11%	3%	13%	159%	-5%	2%	-80%	53%	-10%	-36%	242%	-50%	-27%	-22%	
B	-6%	2%	47%	-76%	2%	8%	-86%	50%	-100%	23%	145%	-50%	-33%	-25%	
C	140%	2%	30%	28%	115%	47%	21%	-86%	-6%	-19%	245%	200%	-19%	-31%	4°C
D	71%	-18%	17%	-23%	53%	20%	-87%	-51%	-22%	-17%	317%	117%	2%	-32%	

OPLS-DA



Loadings of OPLS-DA



Quality Control in NMR analysis

✓ **Robust**
✓ **Quantitative**

